Slodowy Slices and Their Quantization

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Overview

Nilpotent orbits of Lie algebras are important objects to study in representation theory, algebraic geometry, and other related fields. Among them, the Slodowy slice of a nilpotent orbit not only provides numerous examples of algebraic varieties (such as ADE singularities), but also can serve as a source of many important structures (such as W-algebras, Yangians, etc.). In this paper, we study Slodowy slices and their quantization in detail. First, we give the definition of Slodowy slices and define their Poisson structure in two different ways, namely by restricting the given Poisson structure to symplectic leaves and via Hamiltonian reduction. Then we realize the Poisson structure of Slodowy slices by BRST cohomology and quantize them by means of quantized BRST cohomology, thereby obtaining W-algebras. Finally, we present another definition of W-algebras via the Whittaker model and show that it is isomorphic to the previous definition.

Contents

1 Introduction **1**

2 Preliminaries **6**

2.1 Algebraic Geometry 6

2.2 Algebraic Groups 8

2.3 Symplectic Geometry 12

2.4 Clifford Algebras 15

3 Slodowy Slices and Their Poisson Structure **19**

3.1 The Jacobson–Morozov Theorem and the Slodowy Slice 19

3.2 Nilpotent Cone 25

3.3 Poisson Structure on $\mathcal{S}_f$ 30

4 Quantization of the Slodowy Slice and Finite W-Algebras **44**

4.1 BRST Realization of the Slodowy Slice 44

4.2 Quantized BRST Cohomology and Quantization of the Slodowy Slice 52

4.3 Whittaker Model Realization of Finite W-Algebras 56

References **58**

Chapter 1

Introduction

The Slodowy slice is a classic object of study in geometric representation theory, which plays an important role in the study of nilpotent orbits. In the type A case, many significant results have been achieved in the study of Slodowy slices. For example, Maffei showed in [Maf05] that a resolution of a closed subvariety within a Slodowy slice, called a Slodowy variety, is isomorphic to the Nakajima quiver variety $\mathfrak{m}(d, v)$; at the same time, a closed subvariety within the Slodowy slice is isomorphic to the Lagrangian subvariety $\mathfrak{m}^1(d, v)$ of $\mathfrak{m}(d, v)$. Moreover, the following diagram commutes:

$$\begin{tikzcd} \mathfrak{m}(d,v) \arrow[r, "\widetilde{\varphi}"] \arrow[d, "\pi"'] & \widetilde{\mathcal{S}}_{d,x} \arrow[d, "\mu_d"'] \\ \mathfrak{m}^1(d,v) \arrow[r, "\varphi_1"'] & \mathcal{S}_{d,x}. \end{tikzcd}$$

Using Maffei's result, Losev proved in [Los12] that the quantum Hamiltonian reduction associated with type A quiver varieties is isomorphic to the quantization of the type A parabolic Slodowy slice, that is, the type A parabolic W-algebra. However, Maffei's isomorphism is given by a recursive characterization rather than an explicit formula, and similarly, Losev's proof does not write out the isomorphism explicitly.

To address the limitations of the above results, Mirković and Vybornov developed a more explicit geometric method in [MV03], where they used affine Grassmannian slices to construct type A Nakajima quiver varieties. By means of this, they showed that in type A, quiver varieties, nilpotent orbits, and affine Grassmannian singularities are identified. Moreover, in [MV08], they further connected quiver varieties, conjugacy classes of matrices, and Beilinson–Drinfeld Grassmannians, thereby giving a compactification of quiver varieties and decomposing the affine Grassmannian into a disjoint union of quiver varieties. Their construction is in spirit close to Maffei's work, but more explicit; in this sense, it can be viewed as a classical analogue of the Brundan–Kleshchev representation (see [BK06]). It is also closely related to the partly conjectural point of view of symplectic dual pairs (see [Bra+10, Remark 1.5]).

From this perspective, many problems remain to be solved in this field, such as whether similar conclusions hold outside type A; how to study the representations of W-algebras obtained after quantization; how the geometric properties among quiver varieties, Slodowy slices, and affine Grassmannians correspond to one another specifically, etc. Hence, the study of Slodowy slices becomes especially important.

In § 3.1 we define the Slodowy slice following Slodowy's monograph [Slo80]. For a nilpotent element $e \in N$, the celebrated Jacobson–Morozov theorem provides an $\mathfrak{sl}_2$-triple (e, h, f), and we then prove that the affine space

$$S = e + \mathrm{Ker}(\mathrm{ad}\, y)$$

is a special transversal slice to the adjoint orbit through e. We call such a slice a Slodowy slice. Next, in § 3.3 we equip the Slodowy slice with a Poisson structure. By a classic result of Weinstein (see [Wei83]), its Poisson structure can be given by restriction. At the same time, Gan and Ginzburg in [GG02] proposed a structurally richer approach, namely realizing the Slodowy slice as a Poisson variety

via Hamiltonian reduction. Along this direction, the main result of Lehn et al. in [LNS12] gives a necessary and sufficient condition that makes the Slodowy slice isomorphic to the universal Poisson deformation of its nilpotent part.

To further investigate the quantization of the Poisson structure on the Slodowy slice, in § 4.1 we prove that a certain cohomological Poisson algebra is isomorphic to the Poisson structure on the Slodowy slice, thus providing a BRST realization of the Slodowy slice. Immediately after, in § 4.2 we give the quantum version of BRST cohomology, thereby further obtaining a quantization of the Slodowy slice. Finally, in § 4.3 we show that this quantization is precisely the finite W-algebra defined via the Whittaker model (see [Ara07b]). The study of the quantization of Slodowy slices has produced abundant results. For a conic symplectic singularity X, Namikawa proved in [Nam06; Nam10] that the Poisson deformation functor is representable, and Losev showed that the universal Poisson deformation can be quantized and that this quantization possesses a good universal property (see [Los12]). Recently, Ambrosio and Topley in [Amb+23] introduced a quantization functor for flat Poisson deformations and explained that this universal property is equivalent to the representability of the quantization functor; meanwhile, in [AT23] they promoted this picture to the equivariant setting and completely described when a finite W-algebra is the universal filtered quantization of the slice, thereby yielding a nearly complete classification of filtered quantizations of nilpotent Slodowy slices.

The generators and relations of W-algebras have always been a focus of attention. In the finite-dimensional classical W-algebra case, this can be studied via Yangians. The work of Brundan and Kleshchev gave a presentation of the finite W-algebra associated with a nilpotent matrix in the general linear Lie algebra using shifted Yangians (see [BK06]). Inspired by this, Kang Lu et al. in [LWW25] constructed a family of truncated shifted twisted Yangians of type AI, denoted

by $y^+_{n,\ell}(\sigma)$; these algebras are defined by generators and relations and extend the work of Brundan and Kleshchev in a rather different way. They further applied the theory of universal equivariant quantizations of conic symplectic singularities to prove that $y^+_{n,\ell}(\sigma)$ is isomorphic to a finite W-algebra, thereby quantizing the corresponding Slodowy slices of type BCD.

For another classic geometric object, the affine Grassmannian slice, there is a parallel quantization theory. Let G be a complex semisimple group, and let $\mathrm{Gr} = G((t^{-1}))/G[t]$ be the thick affine Grassmannian. For dominant coweights $\lambda \geq \mu$, the transversal slice $\mathrm{Gr}^{\overline{\lambda}}_{\mu}$ is an affine Poisson variety. Kamnitzer et al. in [Kam+14] provided an explicit quantization of its coordinate ring by a quotient of the shifted Yangian. On the other hand, Mirković and Vybornov proved in [MV08] that there exists an isomorphism between the Slodowy slice of $\mathfrak{sl}_m$ and a slice in the affine Grassmannian of GL_n. Thus, quotients of the shifted Yangian of $\mathfrak{gl}_n$ can be viewed as quantizations of affine Grassmannian slices via their relationship with W-algebras. In [WWY20], Webster et al. established a quantized version of the Mirković–Vybornov isomorphism, which connects the intersection of a Slodowy slice with a nilpotent orbit closure in $\mathfrak{gl}_N$ to a slice between spherical Schubert varieties in the affine Grassmannian of PGL_n; the quantization of the former is given by a parabolic W-algebra, and the quantization of the latter by a truncated shifted Yangian. In summary, these results reveal a rich triangular relationship among Nakajima quiver varieties, Slodowy slices, and affine Grassmannian slices, both at the classical and the quantum levels.

Beyond type A, Henderson and Licata showed in [HL14] that, under a diagram automorphism, the fixed-point subvariety of a quiver variety is a disconnected union of quiver varieties of split-quotient quivers. As an application, they obtained isomorphisms between two-row Slodowy slices of types C and D and the corresponding type D quiver varieties, and these isomorphisms are compati-

ble with the natural reductive group actions. In each case, they also established an isomorphism between the slice itself and an affine variety, as well as an isomorphism between the Springer fiber at the base-point of the slice and a projective variety. These works generalize Maffei's results in type A.

It is precisely the abundant achievements and the multitude of open problems in fields related to Slodowy slices that motivate the study of the Slodowy slice in this thesis. In this thesis, we will follow the clues of [Slo80; GG02; AM16]. First, we give the definition of the Slodowy slice and define its Poisson structure in two distinct ways: by restricting the given Poisson structure to symplectic leaves, and via Hamiltonian reduction. Next, we realize the Poisson structure of the Slodowy slice via BRST cohomology, and quantize it via quantized BRST cohomology, thereby obtaining W-algebras. Finally, we present another definition of W-algebras via the Whittaker model and show that it is isomorphic to the previous definition. Meanwhile, since Slodowy slices corresponding to two-row partition-type nilpotent orbits are widely discussed, we will give an example in the case of $\mathfrak{sl}_4$.

Chapter 2

Preliminaries

In this chapter we review some necessary geometric and algebraic knowledge. Specifically, in Section 2.1 we review the definition of the Zarisik tangent space and the differential of relevant morphisms, which is crucial for defining transversal slices; in Section 2.2 we review the definition of semisimple groups and the definitions of exp and log, which will directly serve the actions of algebraic groups on various objects; in Section 2.3 we review the construction of the canonical symplectic structure in symplectic geometry and relevant definitions of Poisson varieties, which will be used frequently in the Hamiltonian reduction process; finally, in Section 2.4 we introduce superalgebras, especially the relevant superalgebra structures on Clifford algebras, a part that contains the core tool for BRST cohomology.

2.1 Algebraic Geometry

This part is covered in many algebraic geometry textbooks; we recommend Hartshorne's [Har77] as a reference.

Definition 2.1. Let X be an algebraic variety over $\mathbb{C}$, and let $x \in X$. We say that

X is *smooth* (also called nonsingular) at x if the local ring $\mathcal{O}_{X,x}$ is a regular local ring. Equivalently,

$$\dim_{\mathbb{C}}(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim X,$$

where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X,x}$. If X is smooth at every point $x \in X$, then X is called a *smooth algebraic variety over* $\mathbb{C}$.

Definition 2.2. Let X be a smooth algebraic variety over $\mathbb{C}$, and let $x \in X$. Denote by $\mathcal{O}_{X,x}$ the local ring at x, and $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ its maximal ideal. The tangent space of X at x is defined as

$$T_xX := \mathrm{Hom}_{\mathbb{C}}\big(\mathfrak{m}_x/\mathfrak{m}_x^2, \mathbb{C}\big).$$

Example 2.3. A finite-dimensional linear space V as an affine algebraic variety is smooth, and its tangent space is isomorphic to itself.

Definition 2.4. If $f : X \to Y$ is a morphism between smooth algebraic varieties and $x \in X$, then the differential is

$$df_x : T_xX \to T_{f(x)}Y.$$

Its action on derivations is given by

$$(df_x(\xi))(g) := \xi(g \circ f), \qquad \xi \in T_xX, \quad g \in \mathcal{O}_{Y,f(x)}.$$

Definition 2.5. Let $f\colon X \to Y$ be a morphism between smooth algebraic varieties. A point $y \in Y$ is called a regular value of f if for every $x \in f^{-1}(y)$, the differential

$$df_x\colon T_xX \to T_yY$$

is surjective.

2.2 Algebraic Groups

In this section we refer to [Spr98; Hum75].

Definition 2.6. An *algebraic group* G is a group that is also an algebraic variety, such that the multiplication map $\mu : G \times G \to G$, where $\mu(x, y) = xy$, and the inversion map $i : x \mapsto x^{-1}$, are morphisms of algebraic varieties, i.e., regular functions. (We may view the underlying variety $G \times G$ as the product of point sets [Spr98, Exercise 1.5.5].) If the underlying variety is affine, then G is called a *linear algebraic group*.

The following theorem explains the origin of the name linear algebraic group:

Proposition 2.7 ([Spr98, Theorem 2.3.7])**.** *A linear algebraic group G is always isomorphic to some closed subgroup of some GL_n.*

Definition 2.8. For an algebraic group G, its Lie algebra is defined as T_eG, where e is the identity element.

Henceforth, unless mentioned otherwise, we will assume all groups G are linear algebraic groups. Any linear algebraic group G over an algebraically closed field k contains a unique maximal connected normal solvable subgroup, called the *radical* of G, denoted by $R(G)$; the subgroup consisting of all unipotent elements in $R(G)$ is normal and closed in G, called the *unipotent radical*, denoted by $R_u(G)$ (see [Spr98, p. 112]).

Definition 2.9. If $R_u(G)$ is trivial and $G \neq e$ is connected, we call G a *reductive group*.

Definition 2.10. If $R(G)$ is trivial and $G \neq e$ is connected, we call G a *semisimple group*.

Remark 2.11. Semisimple groups are reductive.

Example 2.12. The linear algebraic group GL_n is reductive but not semisimple, and its Lie algebra is $\mathfrak{gl}_n$; however, its closed subgroup SL_n is semisimple, and its Lie algebra is $\mathfrak{sl}_n$.

Definition 2.13. Let V be a finite-dimensional vector space.

1. If $n \in \operatorname{End} V$ is nilpotent, then
$$\exp(n) = \sum_{i=0}^{\infty} \frac{n^i}{i!}$$
is a finite sum, which defines an element of $GL(V)$, and $\exp(n)$ is unipotent.

2. If $x \in GL(V)$ is unipotent, then $x - 1$ is nilpotent, and
$$\log(x) = \sum_{i=1}^{\infty} \frac{(-1)^{i+1}}{i}(x-1)^i$$
is also a finite sum, which defines a nilpotent element of $\operatorname{End}(V)$.

Proposition 2.14 ([Hum75, § 15.5 Exercise 8])**.** *In the above definition,* exp *and* log *are mutually inverse bijections between the set of nilpotent matrices in $M(n,k)$ and the set of unipotent matrices in $GL(n,k)$.*

From now on, unless mentioned otherwise, we always assume $k = \mathbb{C}$. For any linear algebraic group G, consider an isomorphic embedding
$$h : \ G \to GL(V),$$
and differentiate at the identity to get
$$dh : \ \mathfrak{g} \to \operatorname{End}(V).$$
We can then define exp and log on G as follows:
$$\exp(g) := h^{-1} \exp(dh(g)) \quad \forall g \in \mathfrak{g};$$
$$\log(x) := dh^{-1} \log(h(x)) \quad \forall x \in G.$$

Remark 2.15. We need to verify that $\exp(dh(g)) \in h(G)$ and $\log(h(x)) \in dh(\mathfrak{g})$. Here we only provide the proof for the former.

Proof. Set $Y := dh(X) \in \mathrm{End}(V)$. Since Y is nilpotent, the map

$$\lambda_X : \mathbb{A}^1 \to GL(V), \qquad t \mapsto \exp(tY)$$

is an algebraic morphism. To prove $\exp(Y) \in h(G)$, it suffices to show $\lambda_X(\mathbb{A}^1) \subset h(G)$.

Let $I(h(G)) \subset k[GL(V)]$ be the defining ideal of $h(G)$. For any $f \in I(h(G))$, set

$$p_f(t) := f(\exp(tY)).$$

Then $p_f(t)$ is a polynomial in t.

Consider the left-invariant vector field on $GL(V)$ given by Y:

$$\widetilde{Y}_g := gY \in T_g GL(V).$$

Since $Y \in \mathrm{Lie}(h(G)) = T_I(h(G))$, for any $g \in h(G)$, the left translation L_g preserves $h(G)$, so

$$\widetilde{Y}_g = d(L_g)_e(Y) \in T_g(h(G)).$$

Thus

$$\widetilde{Y}(I(h(G))) \subset I(h(G)).$$

On the other hand,

$$p_f'(t) = (\widetilde{Y} f)(\exp(tY)),$$

and by induction,

$$p_f^{(m)}(t) = \big(\widetilde{Y}^m f\big)(\exp(tY)).$$

In particular,

$$p_f^{(m)}(0) = \big(\widetilde{Y}^m f\big)(e).$$

Since $f \in I(h(G))$ and $\widetilde{Y}(I(h(G))) \subset I(h(G))$, we obtain

$$\widetilde{Y}^m f \in I(h(G)) \qquad (\forall m \geq 0),$$

and therefore

$$p_f^{(m)}(0) = 0 \qquad (\forall m \geq 0).$$

Since p_f is a polynomial, it follows that $p_f \equiv 0$. Hence

$$f(\exp(tY)) = 0 \qquad \forall t,\ \forall f \in I(h(G)).$$

Consequently,

$$\exp(tY) \in h(G) \qquad \forall t.$$

Taking $t = 1$ in particular yields

$$\exp(dh(X)) = \exp(Y) \in h(G).$$

□

It is easy to see that in this case exp and log are mutually inverse and give a bijection between the unipotent group and the nilpotent Lie algebra.

Example 2.16. Let G be a connected algebraic group with Lie algebra $\mathfrak{g}$. Consider the adjoint orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g)f.$$

Since $\mathfrak{g}$ is a vector space, we make the identification

$$T_f\mathfrak{g} \cong \mathfrak{g}.$$

Also,

$$T_1 G \cong \mathfrak{g}.$$

For $x \in \mathfrak{g}$, take the one-parameter subgroup $g(t) = \exp(tx)$. Then

$$d\mu_f|_1(x) = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}(\exp(tx))f = \frac{d}{dt}\bigg|_{t=0} e^{t\,\mathrm{ad}\,x}(f) = [x, f].$$

Therefore the tangent space of the adjoint orbit $G \cdot f$ at f is

$$T_f(G \cdot f) = \mathrm{Im}(d\mu_f|_1) = [\mathfrak{g}, f] = (\mathrm{ad}\, f)(\mathfrak{g}).$$

2.3 Symplectic Geometry

In this section we mainly refer to [CG10]. Let M be a closed subgroup of a semisimple algebraic group G. We first introduce the coadjoint action. The conjugation action of M on G is defined by

$$\mathrm{Int} : M \to \mathrm{Aut}(G), \qquad m \mapsto \mathrm{Int}_m,$$

where

$$\mathrm{Int}_m(g') = mg'm^{-1}.$$

Differentiating Int_m at the identity yields the adjoint action

$$\mathrm{Ad}_m : \mathfrak{g} \to \mathfrak{g}, \qquad g \mapsto \frac{d}{dt}\bigg|_{t=0} \big(m \exp(tg) m^{-1}\big).$$

Taking its dual map gives

$$\mathrm{Ad}_m^* : \mathfrak{g}^* \to \mathfrak{g}^*.$$

Thus M acts on $\mathfrak{g}^*$ via the coadjoint action:

$$M \to \mathrm{GL}(\mathfrak{g}^*), \qquad m \mapsto \mathrm{Ad}_m^*.$$

Differentiating again at the identity yields

$$\mathfrak{m} \to \mathrm{End}(\mathfrak{g}^*), \qquad m' \mapsto \mathrm{ad}_{m'}^*.$$

Proposition 2.17 (*Kirillov–Kostant–Souriau* symplectic structure, see [CG10, § 1]). *Every coadjoint orbit $\mathcal{O} \subset \mathfrak{g}^*$ admits a natural symplectic structure as follows.*

Take a point $\alpha \in \mathcal{O} \subset \mathfrak{g}^*$. On $T_\alpha \mathcal{O} \simeq \mathfrak{g}/\mathfrak{g}^\alpha$, we have a skew-symmetric form, where $\mathfrak{g}^\alpha = \operatorname{Lie} G^\alpha$. We first define a skew-symmetric form

$$\omega_\alpha \colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad \omega_\alpha(x, y) \mapsto \alpha([x, y]).$$

It descends to $\mathfrak{g}/\mathfrak{g}^\alpha$, i.e.,

$$\omega_\alpha(x, y) = 0 \quad \forall\, y \in \mathfrak{g} \iff x \in \mathfrak{g}^\alpha.$$

One can prove that ω_α is also nondegenerate, so ω_α defines a symplectic form on $\mathcal{O}$. The assignment $\alpha \mapsto \omega_\alpha$ yields a closed 2-form ω on $\mathcal{O}$, which is a symplectic form.

Let A be a commutative, associative, unital $\mathbb{C}$-algebra with multiplication $\cdot \colon A \times A \to A$.

Definition 2.18. A commutative algebra $(A, \cdot)$ equipped with an additional $\mathbb{C}$-bilinear skew-symmetric bracket $\{\,,\,\} \colon A \times A \to A$ is called a *Poisson algebra* if the following hold:

1. A is a Lie algebra with respect to $\{\,,\,\}$;

2. Leibniz rule:

$$\{a, b \cdot c\} = \{a, b\} \cdot c + b \cdot \{a, c\}, \qquad \forall a, b, c \in A.$$

The Lie bracket $\{\,,\,\}$ is called a *Poisson bracket* on A. We say that $\{\,,\,\}$ endows the commutative algebra $(A, \cdot)$ with a *Poisson structure*.

We will derive a Poisson structure called the Lie–Poisson bracket from a symplectic structure. Let (M, ω) be a symplectic variety. The nondegenerate 2-form ω gives a canonical isomorphism $TM \simeq T^*M$. Define the unique $\mathbb{C}$-linear map

$$\mathcal{O}(M) \to \{\text{vector fields on } M\}, \quad f \mapsto \xi_f,$$

such that

$$\omega(\,\cdot\,, \xi_f) = df.$$

Note that for any vector field η and any function f, by the definition of ξ_f we have

$$\omega(\xi_f, \eta) = -\eta f.$$

We define a bracket on $\mathcal{O}(M)$ by any of the following equivalent expressions:

$$\{f, g\} = \omega(\xi_f, \xi_g) = -\xi_g f = \xi_f g.$$

Definition 2.19 (Reduced scheme)**.** A scheme $(X, \mathcal{O}_X)$ is called *reduced* if for every open set $U \subseteq X$, $\mathcal{O}_X(U)$ has no nilpotent elements.

Definition 2.20. An *affine Poisson scheme* (resp. *affine Poisson variety*) is an affine scheme $X = \operatorname{Spec} A$ (resp. $= \operatorname{Specm} A$), where A is a Poisson algebra. A *Poisson scheme* (resp. *Poisson variety*) is a scheme (resp. reduced scheme) whose structure sheaf $\mathcal{O}_X$ is a sheaf of Poisson algebras.

Example 2.21. Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. Then its dual space $\mathfrak{g}^*$ is naturally a Poisson variety. Indeed, as an affine space,

$$\mathfrak{g}^* = \operatorname{Spec} \mathbb{C}[\mathfrak{g}^*],$$

and

$$\mathbb{C}[\mathfrak{g}^*] \cong S(\mathfrak{g}).$$

For $x, y \in \mathfrak{g} \subset S(\mathfrak{g})$, define

$$\{x, y\} := [x, y]_{\mathfrak{g}},$$

and extend it uniquely to the whole $S(\mathfrak{g})$ by the Leibniz rule. Then $S(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*]$ becomes a Poisson algebra. Hence $\mathfrak{g}^*$ is an affine Poisson variety.

Definition 2.22. A Hamiltonian flow from x to x' is a smooth curve $\gamma\colon [0,1] \to X$ in X, defined on an open neighborhood of $[0,1]$ in $\mathbb{C}$, satisfying $\gamma(0) = x$, $\gamma(1) = x'$, and γ is an integral curve of some Hamiltonian vector field $\xi_f = \{f, \cdot\}$ with $f \in \mathcal{O}(X)$ defined on an open neighborhood of $\gamma([0,1])$.

2.4 Clifford Algebras

This section is mainly taken from [AM24; Mei13]. We will introduce superalgebras, in particular Clifford algebras.

Definition 2.23. A superspace is a $\mathbb{Z}_2$-graded $\mathbb{C}$-linear space E, i.e., $E = E_{\bar{0}} \oplus E_{\bar{1}}$. Elements in $E_{\bar{0}}$ are called *even*, and elements in $E_{\bar{1}}$ are called *odd*. A superspace which is also an algebra is called a *superalgebra*.

Definition 2.24. Let $\mathcal{A} = \mathcal{A}_{\bar{0}} \oplus \mathcal{A}_{\bar{1}}$ be a superspace. If a bilinear map

$$[\cdot,\cdot] : \mathcal{A} \times \mathcal{A} \to \mathcal{A},$$

is given such that for all homogeneous elements $x, y, z \in \mathcal{A}$, we have

$$\begin{aligned} [x,y] &= -(-1)^{|x||y|}[y,x], \\ [x,[y,z]] &= -[z,[x,y]] - (-1)^{|x||y|+1}[y,[x,z]] \text{ (Jacobi superidentity)}, \end{aligned}$$

where $|\cdot|$ is the degree operator, then $(\mathcal{A}, [\cdot,\cdot])$ is called a *Lie superalgebra*.

Remark 2.25. Any superalgebra $\mathcal{A}$ can be made into a natural Lie superalgebra by defining

$$[x,y] := xy - (-1)^{|x||y|}yx.$$

If for all homogeneous $x, y \in \mathcal{A}$ we have

$$[x, y] = 0,$$

i.e.,

$$xy = (-1)^{|x||y|} yx,$$

then $\mathcal{A}$ is called *supercommutative*.

Definition 2.26. Let $\mathcal{A} = \mathcal{A}_{\bar{0}} \oplus \mathcal{A}_{\bar{1}}$ be a supercommutative superalgebra. If $\mathcal{A}$ is also equipped with a bilinear map

$$\{\cdot, \cdot\} : \mathcal{A} \times \mathcal{A} \to \mathcal{A},$$

such that:

1. $(\mathcal{A}, \{\cdot, \cdot\})$ is a Lie superalgebra;
2. for all homogeneous $x, y, z \in \mathcal{A}$, the bracket satisfies the super Leibniz rule

$$\{x, yz\} = \{x, y\}z + (-1)^{|x||y|} y\{x, z\},$$

   (i.e., $\{x, \cdot\} : \mathcal{A} \to \mathcal{A}$ is a super derivation),

then $(\mathcal{A}, \cdot, \{\cdot, \cdot\})$ is called a *Poisson superalgebra*.

Example 2.27. Let E be a $\mathbb{C}$-linear space. The *exterior algebra* $\wedge E$ is the quotient of the tensor algebra $T(E) = \bigoplus_{k \in \mathbb{Z}} T(E)^k$ by the two-sided ideal $I(E)$ generated by elements of the form $v \otimes w + w \otimes v$ (with $v, w \in E$), where $T^k(E) = E \otimes \cdots \otimes E$ is the k-th tensor power. Multiplication in $\wedge E$ is usually denoted $v \wedge w$. Since $I(E)$ is graded, the exterior algebra acquires a $\mathbb{Z}$-grading

$$\wedge(E) = \bigoplus_{k \in \mathbb{Z}} \wedge^k(E).$$

We regard $\wedge(E)$ as a graded superalgebra, where the $\mathbb{Z}_2$-grading is obtained from the $\mathbb{Z}$-grading modulo 2.

Definition 2.28. The Clifford algebra $Cl(E, B)$ is the quotient of $T(E)$ by the two-sided ideal $\mathcal{I}(E, B)$ generated by all elements of the form

$$v \otimes w + w \otimes v - B(v, w)1, \quad v, w \in E,$$

where $B : E \times E \to E$ is a symmetric bilinear form (usually required to be nondegenerate).

We regard $T(E) = \bigoplus_{k \in \mathbb{Z}} T^k(E)$ as a filtered superalgebra, where both the $\mathbb{Z}_2$-grading and the filtration are inherited from the $\mathbb{Z}$-grading. Since the elements $v \otimes w + w \otimes v - B(v, w)1$ are even and have filtration degree 2, the ideal $\mathcal{I}(E, B)$ is a filtered super subspace of $T(E)$, and therefore $Cl(E, B)$ inherits a filtered superalgebra structure. In the decomposition

$$Cl(E, B) = Cl(E, B)^{\bar{0}} \oplus Cl(E, B)^{\bar{1}},$$

the two direct summands are spanned respectively by products $v_1 \dots v_k$ with k even or odd. We always view $Cl(E, B)$ as a filtered superalgebra. Then the defining relations of the Clifford algebra become

$$[v, w] = vw + wv = B(v, w), \qquad v, w \in E.$$

We define the map

$$q\colon \wedge(E) \to Cl(E, B), \quad v_1 \wedge \dots \wedge v_k \mapsto \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma) v_{\sigma 1} \dots v_{\sigma k}, \tag{2.i}$$

where $\mathfrak{S}_k$ is the k-th permutation group. This map is an isomorphism of superspaces. Its inverse is called the *symbol map*. From [Mei13, § 2.2.5] we have

Proposition 2.29. *The symbol map $\sigma\colon Cl(E, B) \to \wedge(E)$ induces an isomorphism of graded superalgebras,*

$$\operatorname{gr} Cl(E, B) \xrightarrow{\sim} \wedge(E).$$

$\operatorname{gr} Cl(E,B)$ inherits a Poisson superalgebra structure, and the symbol map is an isomorphism of graded Poisson superalgebras. The Poisson bracket on $\wedge(E)$ can be described as:

$$\{v,w\} = B(v,w), \quad v,w \in E = \wedge^1(E).$$

Further, let $(\mathcal{A}, \{,\}_{\mathcal{A}})$ and $(\mathcal{B}, \{,\}_{\mathcal{B}})$ be two Poisson algebras. Consider the tensor product of associative algebras $\mathcal{A} \otimes_{\mathbb{C}} \mathcal{B}$ and endow it with the following Poisson bracket:

$$\{x_1 \otimes y_1, x_2 \otimes y_2\} = \{x_1, x_2\}_A \otimes y_1 \cdot y_2 + x_1 \cdot x_2 \otimes \{y_1, y_2\}_B,$$

for $x_1, x_2 \in A$, $y_1, y_2 \in B$. It is easy to verify that $A \otimes_{\mathbb{C}} B$ is a Poisson algebra. Moreover, since $1 \otimes B$ is isomorphic to B as Poisson algebras, we may write $B \subset A \otimes_{\mathbb{C}} B$ (using the fact that $\{1, x\} = 0$ for all $x \in A$).

This observation can be readily generalized to the case of Clifford algebras, where the Poisson bracket becomes

$$\{x_1 \otimes y_1, x_2 \otimes y_2\} = (-1)^{|x_2||y_1|}\{x_1, x_2\}_A \otimes y_1 \cdot y_2 + (-1)^{|x_2|(|x_1|+|y_1|)} x_1 \cdot x_2 \otimes \{y_1, y_2\}_B.$$

Chapter 3

Slodowy Slices and Their Poisson Structure

In this chapter we will investigate in depth the Poisson structure on Slodowy slices. In Section 3.1 we use the results of Jacobson and Morozov to define the Slodowy slice and prove its transversality; in Section 3.2 we study simple properties of the nilpotent cone; in Section 3.3 we give the realization of the Slodowy slice on the dual space and endow it with a Poisson structure in two ways, namely by restricting the given Poisson structure to symplectic leaves and by Hamiltonian reduction.

3.1 The Jacobson–Morozov Theorem and the Slodowy Slice

In this section we mainly follow the notations of [Slo80].

Let $\mathfrak{sl}_2 = kE + kH + kF$, with elements

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

satisfying the commutation relations:

$$[H, E] = 2E,\ [H, F] = -2F,\ [E, F] = H.$$

Definition 3.1. For a nilpotent element e in $\mathfrak{g}$, if there exist $h, f \in \mathfrak{g}$ and a Lie algebra homomorphism $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$ such that $\rho(E) = e$, $\rho(H) = h$, $\rho(F) = f$, then (e, h, f) is called an $\mathfrak{sl}_2$-triple for the nilpotent element e.

Definition 3.2. A representation $\rho : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ of $\mathfrak{sl}_2$ on a finite-dimensional vector space V is called *good* if either $\mathrm{char}(k) = 0$, or $\mathrm{char}(k) = p > 0$ and $\rho(E)^{p-1} = \rho(F)^{p-1} = 0$.

Let the d-dimensional vector space V_d be spanned by basis vectors $v_1, \ldots, v_d$. If $\mathrm{char}(k) = p > 0$, we require $d < p$, otherwise no restriction on d. There exists a *good*, d-dimensional, irreducible $\mathfrak{sl}_2$-representation defined by:

$$\begin{aligned}
\rho_d(E)v_1 &= \rho_d(F)v_d = 0 \\
\rho_d(E)v_{i+1} &= \overline{i(d-i)}\, v_i \quad \text{for } 1 \le i \le d-1 \\
\rho_d(F)v_i &= v_{i+1} \quad \text{for } 1 \le i \le d-1 \\
\rho_d(H)v_i &= \overline{(d-2i+1)}\, v_i
\end{aligned}$$

(where for $n \in \mathbb{Z}$, $\overline{n}$ denotes the residue class modulo p.)

The following theorem is key to the definition of the Slodowy slice. Let $\mathfrak{g}$ be the Lie algebra of a semisimple algebraic group G (of type Δ)[1], and let $x \in \mathcal{N}(\mathfrak{g})$ be a nilpotent element in $\mathfrak{g}$. The *Coxeter number* $\mathrm{Cox}(G)$ of G or $\mathrm{Cox}(\Delta)$ of Δ is defined as $m + 1$, where m is the height of the highest root in the root system of G. The values of $\mathrm{Cox}(G)$ are as follows (see [Slo80, § 7]):

Next we have the celebrated Jacobson–Morozov theorem (see [Slo80, § 7.2]):

[1] It is known that $\mathfrak{g}$ is semisimple, see [Hum75, § 13.5]

type of G	A_r	B_r	C_r	D_r	E_6	E_7	E_8	F_4	G_2
$\mathrm{Cox}(G)$	$r+1$	$2r$	$2r$	$2r-2$	12	18	30	12	6

Theorem 3.3 (Jacobson–Morozov)**.** *Assume* $\mathrm{char}(k) = 0$ *or* $> 4\,\mathrm{Cox}(G) - 2$*. For every nilpotent element* $e \in \mathfrak{g}$*, there exists an* $\mathfrak{sl}_2$*-triple, and the corresponding representation* $\rho : \mathfrak{sl}_2 \to \mathfrak{g}$ *composed with the adjoint representation* $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ *is* good.

In this thesis, we mainly focus on the Lie algebra $\mathfrak{sl}_n$ corresponding to the group SL_n. Every nilpotent matrix in $\mathfrak{sl}_n$ is conjugate to a block-diagonal Jordan matrix. Hence the nilpotent orbits in $\mathfrak{g}$ are parameterized by $\mathcal{P}(n)$. Consequently, nilpotent orbits in $\mathfrak{sl}_n$ correspond to partitions of n.

Remark 3.4. Let $n \in \mathbb{Z}_{>0}$. An element of a partition of n, denoted $\boldsymbol{\lambda} \in \mathcal{P}(n)$, is written as a decreasing sequence $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_s)$, omitting zeros. Thus

$$\lambda_1 \geq \cdots \geq \lambda_s \geq 1 \quad \text{and} \quad \lambda_1 + \cdots + \lambda_s = n.$$

We denote the dual partition of $\boldsymbol{\lambda}$ by ${}^t\boldsymbol{\lambda}$, whose Young tableau is the transpose of that of $\boldsymbol{\lambda}$.

Example 3.5. Let $\mathfrak{g} = \mathfrak{sl}_n$, and set

$$e := J_n, \qquad h := \sum_{i=1}^{n} (n+1-2i) e_{i,i}, \qquad f := \sum_{i=1}^{n-1} i(n-i) e_{i+1,i}.$$

Then (e, h, f) is an $\mathfrak{sl}_2$-triple. From this observation, we can immediately construct an $\mathfrak{sl}_2$-triple for any Jordan canonical form $\mathrm{diag}(J_{\lambda_1}, \ldots, J_{\lambda_n})$ where $(\lambda_1, \ldots, \lambda_n) \in \mathcal{P}(n)$. For the partition $(3, 1)$, we obtain

$$e = E_{12} + E_{23}, \qquad h = \mathrm{diag}(2, 0, -2, 0), \qquad f = 2E_{21} + 2E_{32}.$$

One can directly check

$$[h, e] = 2e, \qquad [e, f] = h, \qquad [h, f] = -2f.$$

Next we state a theorem proved by Jacobson in [Jac58]:

Theorem 3.6. *Every good representation ρ is completely reducible and decomposes into a direct sum of irreducible good representations of type ρ_d, where $d \in \mathbb{N}^+$, and if* $\mathrm{char}(k) = p > 0$ *then $d < p$. In particular, for each natural number $d \in \mathbb{N}^+$, with the additional condition $d < p$ if* $\mathrm{char}(k) = p > 0$*, there exists exactly one irreducible representation of dimension d.*

We can now discuss the weight space decomposition over more general fields satisfying the conditions. On the representation space V_d of the good representation ρ_d, we can define an action of the multiplicative group k^* by linearly extending the following action:

$$(t, v_i) \mapsto t^{d-2i+1} \cdot v_i, \quad t \in k^*,\ 1 \leq i \leq d.$$

If $\rho : \mathfrak{sl}_2 \to \mathfrak{gl}(V)$ is any good representation, we can define a unique k^*-action $\lambda : k^* \to \mathrm{GL}(V)$ on V by letting k^* act on each irreducible part of type V_d as above. Differentiating λ, $D_e\lambda : \mathrm{Lie}(k^*) \to \mathfrak{gl}(V)$, gives $D_e(\lambda)(1) = \rho(H)$ (identifying $\mathrm{Lie}(k^*)$ with k). If a vector $v \in V$ in a good representation satisfies

$$\lambda(t)v = t^n v \quad \forall t \in k^*,$$

it is called an eigenvector of *integral weight* $n \in \mathbb{Z}$ for λ (or H). When $\mathrm{char}(k) = 0$, this condition is equivalent to the infinitesimal condition

$$\rho(H)v = nv.$$

In what follows, we always understand the weights of H as integral weights. For any $n \in \mathbb{Z}$, set

$$V(n) = \{v \in V \mid \lambda(t)v = t^n v\},$$

called the weight space of weight n. Let $V = V_d$ be the vector space of the good representation ρ_d. Then V decomposes into a direct sum of d one-dimensional

weight spaces $V(n)$, where $n = d-1, d-3, \ldots, -(d-3), -(d-1)$. The nilpotent inner derivations $\rho_d(E)$ (resp. $\rho_d(F)$) induce isomorphisms

$$E : V(n) \xrightarrow{\sim} V(n+2) \quad \text{when } n < d-1$$

(resp. $F : V(n) \xrightarrow{\sim} V(n-2)$ when $n > -(d-1)$), and annihilate $V(d-1)$ (resp. $V(-(d-1))$).

Definition 3.7. Let V be a smooth algebraic variety, $\mathcal{O} \subset V$ a smooth subvariety, and $y \in \mathcal{O}$. A subvariety $S \subset V$ passing through y is called a *transversal slice to $\mathcal{O}$ at y* if

$$T_y V = T_y \mathcal{O} \oplus T_y S.$$

Let (e, h, f) be an $\mathfrak{sl}_2$-triple in $\mathfrak{g}$. Consider the orbit map

$$\mu_f : G \to \mathfrak{g}, \qquad g \mapsto \mathrm{Ad}(g) f.$$

By Example 2.16, the tangent space of the adjoint orbit $G \cdot f$ at f is

$$T_f(G \cdot f) = (\mathrm{ad}\, f)(\mathfrak{g}).$$

By Theorem 3.6, the adjoint representation of the $\mathfrak{sl}_2$-triple decomposes as

$$\mathfrak{g} = \bigoplus_{j=1}^{s} E_j,$$

where each E_j is an irreducible $\mathfrak{sl}_2$-module of dimension $n_j + 1$. Write

$$E_j = \bigoplus_{m=-n_j}^{n_j} E_j(m)$$

for its weight space decomposition. Each $E_j(m)$ is one-dimensional, and for every $m \leq n_j - 2$, the map $\mathrm{ad}\, f$ restricts to an isomorphism

$$\mathrm{ad}\, f : E_j(m+2) \xrightarrow{\sim} E_j(m).$$

Therefore,

$$(\operatorname{ad} f)(E_j) = \bigoplus_{m=-n_j}^{n_j-2} E_j(m),$$

so the highest weight space $E_j(n_j)$ is a complement of $(\operatorname{ad} f)(E_j)$ in E_j. Since

$$E_j(n_j) = \ker(\operatorname{ad} e) \cap E_j,$$

we obtain

$$\ker(\operatorname{ad} e) = \bigoplus_{j=1}^{s} E_j(n_j).$$

Hence,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e).$$

Definition 3.8. The Slodowy slice through f is the affine subspace

$$\mathcal{S}_f := f + \ker(\operatorname{ad} e) = f + \mathfrak{g}^e,$$

Proposition 3.9. *The Slodowy slice $\mathcal{S}_f$ is transverse to the adjoint orbit $G \cdot f$ at the point f. More precisely,*

$$T_f\mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Proof. Since $\mathcal{S}_f = f + \mathfrak{g}^e$ is an affine subspace of $\mathfrak{g}$, its tangent space at f is

$$T_f(\mathcal{S}_f) = \mathfrak{g}^e = \ker(\operatorname{ad} e).$$

On the other hand,

$$T_f(G \cdot f) = (\operatorname{ad} f)(\mathfrak{g}).$$

Using the decomposition above,

$$\mathfrak{g} = (\operatorname{ad} f)(\mathfrak{g}) \oplus \ker(\operatorname{ad} e),$$

we get

$$T_f\mathfrak{g} = \mathfrak{g} = T_f(G \cdot f) \oplus T_f(\mathcal{S}_f).$$

Thus $\mathcal{S}_f$ is transverse to the orbit $G \cdot f$ at f. $\square$

In particular, $\mathcal{S}_f$ defines a transversal slice to the adjoint orbit of f at the point f.

Example 3.10. Let $Y = (y_{ij}) \in \mathfrak{sl}_4$. Solve the equation

$$[Y, e] = 0, \qquad e = E_{12} + E_{23},$$

to obtain

$$Y = \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix}.$$

Therefore

$$\mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 0 & a & b & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \; a, b, c, u, v \in \mathbb{C} \right\}.$$

Thus

$$\mathcal{S}_f = f + \mathfrak{g}^e = \left\{ \begin{pmatrix} a & b & c & u \\ 2 & a & b & 0 \\ 0 & 2 & a & 0 \\ 0 & 0 & v & -3a \end{pmatrix} \middle| \; a, b, c, u, v \in \mathbb{C} \right\}.$$

Hence, as an affine algebraic variety,

$$\mathcal{S}_f \cong \mathbb{A}^5$$

with coordinate ring

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[a, b, c, u, v].$$

3.2 Nilpotent Cone

In this section we mainly refer to [AM16].

Definition 3.11. The set of all nilpotent elements in a Lie algebra $\mathfrak{g}$ is called the *nilpotent cone*, denoted $\mathcal{N}$.

For $e \in \mathfrak{g}$, we denote its G-orbit under the adjoint action by $G.e$. The nilpotent cone is a union of finitely many nilpotent G-orbits. An element $x \in \mathfrak{g}$ is called *regular* if its centralizer $\mathfrak{g}^x$ has the minimal dimension, namely the rank r of $\mathfrak{g}$. The orbit of maximal dimension $\mathcal{O}_{\text{reg}}$ is the set of all regular nilpotent elements.

Note that if $\mathfrak{g}$ is a simple Lie algebra of matrices, then due to the properties of Jordan decomposition, $\mathcal{N}$ coincides with the set of nilpotent matrices (see [Hum72, § 6.3]).

Example 3.12. In $\mathfrak{sl}_n$, a regular nilpotent element is the conjugacy class of the nilpotent Jordan block of size n:

$$e = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

The matrices in $\mathfrak{gl}_n$ that commute with e are polynomials in e:

$$a_0 I + a_1 e + \cdots + a_{n-1} e^{n-1},$$

so their dimension is n. In $\mathfrak{sl}_n$, taking the trace-zero quotient eliminates the generator I. Hence

$$\dim \mathfrak{g}^e = n - 1 = \operatorname{rank} \mathfrak{sl}_n.$$

Next, within $\mathcal{N} \setminus \mathcal{O}_{\text{reg}}$, there exists a unique orbit of maximal dimension, called the *subregular nilpotent orbit* of $\mathfrak{g}$, denoted $\mathcal{O}_{\text{subreg}}$.

Example 3.13. Consider the nilpotent Jordan block with partition $(n-1, 1)$, i.e., one Jordan block of size $(n-1)$ and one of size 1, denoted X.

$$\mathfrak{g}^X = \{Y \in \mathfrak{sl}_n : [X, Y] = 0\}.$$

Write

$$Y = \begin{pmatrix} A & b \\ c & d \end{pmatrix},$$

with $A \in M_{n-1}$, $b \in M_{n-1,1}$, $c \in M_{1,n-1}$, $d \in \mathbb{C}$.

$$X = \begin{pmatrix} J & 0 \\ 0 & 0 \end{pmatrix}, \quad J := J_{n-1},$$

where J_{n-1} is the nilpotent Jordan block of size $(n-1)$. Then

$$XY = \begin{pmatrix} JA & Jb \\ 0 & 0 \end{pmatrix}, \quad YX = \begin{pmatrix} AJ & 0 \\ cJ & 0 \end{pmatrix}.$$

So we have

$$JA = AJ, \quad Jb = 0, \quad cJ = 0.$$

Thus

$$c = (0, \ldots, 0, c_{n-1}), \quad b = (b_1, 0, \ldots, 0)^T, \quad A = a_0 I + a_1 J + \cdots + a_{n-2} J^{n-2}.$$

Because $\operatorname{tr} Y = 0$, d is uniquely determined by A. Hence its dimension is uniquely determined by the above, namely $1 + 1 + n - 1 = n - 1 + 2 = \operatorname{rank} \mathfrak{sl}_n + 2$.

In particular, in most parts of this section we focus on the subregular case of $\mathfrak{sl}_4$, i.e., the partition $(3, 1)$ of 4. Furthermore, in examples thereafter we shall not distinguish between $\mathfrak{g}$ and $\mathfrak{sl}_4$.

Example 3.14. We denote the corresponding nilpotent orbit in $\mathfrak{sl}_n$ by $\mathcal{O}_{\boldsymbol{\lambda}}$. Then $\mathcal{O}_{\boldsymbol{\lambda}}$ is represented by the standard Jordan form $\operatorname{diag}(J_{\lambda_1}, \ldots, J_{\lambda_s})$, where J_k is the Jordan block of size k. If we set ${}^t\boldsymbol{\lambda} = (d_1, \ldots, d_t)$, then one can prove that for some $e \in \mathcal{O}_{\boldsymbol{\lambda}}$,

$$\dim \mathfrak{g}^e = \sum_i d_i^2 - 1.$$

Hence

$$\dim \mathcal{O}_{\boldsymbol{\lambda}} = n^2 - \sum_{i=1}^{t} d_i^2.$$

For $\boldsymbol{\lambda} = (3,1)$ in $\mathfrak{sl}_4$, since ${}^t(3,1) = (2,1,1)$, we obtain

$$\dim \mathcal{O}_{(3,1)} = 4^2 - (2^2 + 1^2 + 1^2) = 16 - 6 = 10.$$

Therefore

$$\dim \mathfrak{g}^e = \dim \mathfrak{sl}_4 - \dim \mathcal{O}_{(3,1)} = 15 - 10 = 5.$$

From the Jacobson–Morozov Theorem 3.3 we obtain an $\mathfrak{sl}_2$-triple e, f, h. In particular, we have the following:

Definition 3.15. Since the eigenvalues of ad h are integers, we obtain a $\frac{1}{2}\mathbb{Z}$-grading of $\mathfrak{g}$ defined by h, called the *Dynkin grading associated with* h:

$$\mathfrak{g} = \bigoplus_{j \in \frac{1}{2}\mathbb{Z}} \mathfrak{g}_j, \qquad \mathfrak{g}_j := \{x \in \mathfrak{g} \mid [h, x] = 2jx\}.$$

If $\mathfrak{g}_i = \{0\}$ for all half-integers i, then e is called an *even nilpotent element*. The nilpotent orbit of an even nilpotent element is called an *even nilpotent orbit*.

Example 3.16. Since

$$h = \mathrm{diag}(2, 0, -2, 0),$$

set

$$w_1 = 2, \qquad w_2 = 0, \qquad w_3 = -2, \qquad w_4 = 0,$$

then

$$[h, E_{ij}] = (w_i - w_j)E_{ij}.$$

Hence

$$\mathfrak{g}_2 = \mathbb{C}E_{13},$$

$$\mathfrak{g}_1 = \mathrm{span}\{E_{12}, E_{14}, E_{23}, E_{43}\},$$

$$\mathfrak{g}_0 = \left\{ \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & 0 & u \\ 0 & 0 & c & 0 \\ 0 & v & 0 & d \end{pmatrix} \middle| \; a+b+c+d=0 \right\},$$

$$\mathfrak{g}_{-1} = \operatorname{span}\{E_{21}, E_{41}, E_{32}, E_{34}\}, \qquad \mathfrak{g}_{-2} = \mathbb{C}E_{31}.$$

In particular, this nilpotent element is *even*.

In what follows we will often identify $\mathfrak{g}$ with its dual $\mathfrak{g}^*$ via a nondegenerate invariant bilinear form $(\ |\ )$; in particular, in the semisimple type A case, this nondegenerate bilinear form can be taken to be the Killing form. Let (e, h, f) be an $\mathfrak{sl}_2$-triple of $\mathfrak{g}$, and set $\chi = (f|\cdot)$ to be the linear functional associated with f. Then we have

Proposition 3.17. *The skew-symmetric bilinear form*

$$\omega_\chi : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}, \qquad (x, y) \mapsto (f|[x, y]),$$

restricted to $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{\frac{1}{2}}$ *is nondegenerate.*

Proof. It is easy to see that $\operatorname{ad} f : \mathfrak{g}_{\frac{1}{2}} \to \mathfrak{g}_{-\frac{1}{2}}$ is a bijection with inverse $\operatorname{ad} e$. Hence

$$\omega_\chi(x, y) = (f \mid [x, y]) = ([f, x] \mid y),$$

because the bilinear form is invariant under the Lie bracket. Using the fact that $(\ |\ )$ is nondegenerate on $\mathfrak{g}_{\frac{1}{2}} \times \mathfrak{g}_{-\frac{1}{2}}$, if $\omega_\chi(x, y) = 0$ for all y, then since the above $\operatorname{ad} f$ is injective, we have $[f, x] = 0$, which implies $x = 0$. Hence ω_χ is nondegenerate on $\mathfrak{g}_{\frac{1}{2}}$. $\square$

We call ω_χ the *Kirillov form associated with* f. Let $\mathfrak{L}$ be a Lagrangian subspace of $\mathfrak{g}_{\frac{1}{2}}$, i.e., $\omega_\chi(\mathfrak{L}, \mathfrak{L}) = 0$ and $\dim \mathfrak{L} = \frac{1}{2} \dim \mathfrak{g}_{\frac{1}{2}}$, and set

$$\mathfrak{m} = \mathfrak{m}_{\chi,\mathfrak{L}} := \mathfrak{L} \oplus \bigoplus_{j > \frac{1}{2}} \mathfrak{g}_j. \tag{3.i}$$

Then $\mathfrak{m}$ is an ad-nilpotent (since $[\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$), $\operatorname{ad} h$-graded subalgebra of $\mathfrak{g}$. Moreover, because $(\mathfrak{g}_i \mid \mathfrak{g}_j) = 0$ unless $i + j = 0$, we have $\chi([\mathfrak{m}, \mathfrak{m}]) = (f|[\mathfrak{m}, \mathfrak{m}]) = 0$, i.e., χ is a character on $\mathfrak{m}$.

Example 3.18. In our case

$$\chi = (f \mid \cdot), \qquad f = 2E_{21} + 2E_{32}.$$

Due to the well-known fact $k(A, B) := \operatorname{tr}(\operatorname{ad} A \cdot \operatorname{ad} B) = 2n \operatorname{tr}(AB)$, we only need to compute the trace pairing $\chi := tr(A\cdot)$,

$$\chi(E_{12}) = 2, \qquad \chi(E_{23}) = 2, \qquad \chi(E_{14}) = 0, \qquad \chi(E_{43}) = 0, \qquad \chi(E_{13}) = 0.$$

And because

$$\mathfrak{g}_{1/2} = 0,$$

we have $\mathfrak{L} = 0$, thus

$$\mathfrak{m} = \mathfrak{g}_1 \oplus \mathfrak{g}_2.$$

Therefore

$$\mathfrak{m} = \operatorname{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\}.$$

3.3 Poisson Structure on $\mathcal{S}_f$

In this section we mainly refer to [AM16; GG02]. First, we consider the realization of the Slodowy slice on the dual space $\mathfrak{g}^*$. Set

$$\chi := (f \mid \cdot) \in \mathfrak{g}^*,$$

and define

$$\mathcal{S}_f := \chi + (\mathfrak{g}^f)^* \subset \mathfrak{g}^*.$$

We now show that, via an invariant bilinear form, this affine subspace is naturally identified with the original definition

$$\mathcal{S}_f = f + \mathfrak{g}^e.$$

Proposition 3.19. *Let*

$$\kappa : \mathfrak{g} \xrightarrow{\sim} \mathfrak{g}^*, \qquad x \mapsto (x \mid \cdot)$$

be the isomorphism induced by the invariant bilinear form. Then

$$\kappa\big(f + \mathfrak{g}^e\big) = \chi + (\mathfrak{g}^f)^*.$$

Equivalently,

$$\mathcal{S}_f \cong f + \mathfrak{g}^e.$$

Proof. Since $\kappa(f) = \chi$, it suffices to prove that the restriction of κ induces an isomorphism

$$\mathfrak{g}^e \xrightarrow{\sim} (\mathfrak{g}^f)^*.$$

Consider the linear map

$$\varphi : \mathfrak{g}^e \to (\mathfrak{g}^f)^*, \qquad x \mapsto (x \mid \cdot)\big|_{\mathfrak{g}^f}.$$

We first prove that φ is injective. Suppose $x \in \mathfrak{g}^e$ satisfies

$$(x \mid y) = 0 \qquad \text{for all } y \in \mathfrak{g}^f.$$

By the $\mathfrak{sl}_2$-module decomposition of $\mathfrak{g}$, similar to Definition 3.9, we have

$$\mathfrak{g} = [\mathfrak{g}, f] \oplus \mathfrak{g}^e.$$

On the other hand, by the invariance of $(\cdot \mid \cdot)$, for any $z \in \mathfrak{g}$ and $y \in \mathfrak{g}^f$,

$$([z, f] \mid y) = (z \mid [f, y]) = 0,$$

since $[f, y] = 0$. Hence

$$[\mathfrak{g}, f] \subseteq (\mathfrak{g}^f)^\perp.$$

Now

$$\dim[\mathfrak{g}, f] = \dim \mathfrak{g} - \dim \mathfrak{g}^f = \dim(\mathfrak{g}^f)^\perp,$$

so

$$[\mathfrak{g}, f] = (\mathfrak{g}^f)^\perp.$$

Thus $x \in \mathfrak{g}^e \cap (\mathfrak{g}^f)^\perp = \mathfrak{g}^e \cap [\mathfrak{g}, f]$. Since

$$\mathfrak{g} = [\mathfrak{g}, f] \oplus \mathfrak{g}^e,$$

this intersection is zero, so $x = 0$. Therefore φ is injective.

Now compare dimensions. We have proved in Definition 3.9 that $\ker(\operatorname{ad} e) = \bigoplus_{j=1}^s E_j(n_j)$, and similarly $\ker(\operatorname{ad} f) = \bigoplus_{j=1}^s E_j(-n_j)$, hence

$$\dim \mathfrak{g}^e = \dim \mathfrak{g}^f.$$

Thus

$$\dim \mathfrak{g}^e = \dim(\mathfrak{g}^f)^*.$$

Since φ is an injective linear map between finite-dimensional vector spaces of the same dimension, it is an isomorphism. Finally, we obtain

$$\kappa\big(f + \mathfrak{g}^e\big) = \chi + (\mathfrak{g}^f)^*.$$

□

If we consider general $\mathfrak{m}$, it is believed that finally the given W-algebra will be the same no matter which $\mathfrak{m}$ to choose. But it's still important to see the definition which we actually will not use later. Let $\mathfrak{L}$ be an isotropic subspace of $\mathfrak{g}_{\frac{1}{2}}$, i.e., $\omega_\chi(\mathfrak{L}, \mathfrak{L}) = 0$, and set

$$\mathfrak{m}_{\chi,\mathfrak{L}} := \mathfrak{L} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j, \qquad \mathfrak{n}_{\chi,\mathfrak{L}} := \mathfrak{L}^{\perp_{\omega_\chi}} \oplus \bigoplus_{j>\frac{1}{2}} \mathfrak{g}_j,$$

where

$$\mathfrak{L}^{\perp_{\omega_\chi}} = \left\{ x \in \mathfrak{g}_{\frac{1}{2}} \mid \omega_\chi(x, \mathfrak{L}) = 0 \right\}.$$

Note that $\mathfrak{g}^e \subset \mathfrak{m}^\perp$, because $\mathfrak{g}^e$ consists of all highest weight vectors; here $\perp$ denotes the orthogonal complement with respect to the bilinear form $(\ |\ )$.

Let $\mathbf{N}$ be the unipotent subgroup of G; thus it is connected (see [Hum75, Exercise 15.6]), and its Lie algebra is $\mathfrak{n}$. When $\mathfrak{m} = \mathfrak{n}$, $\mathfrak{L}$ is Lagrangian, because for any symplectic space (V, ω) we have $\dim L + \dim L^{\perp\omega} = \dim V$. We also denote by $\mathbf{M}$ the closed connected subgroup of G whose Lie algebra is $\mathfrak{m}$ (see [Hum75, § 13.1]).

Consider the adjoint map

$$\mathbf{N} \times (f + \mathfrak{m}^\perp) \longrightarrow \mathfrak{g}, \quad (g, x) \mapsto g \cdot x.$$

Proposition 3.20. *The image of the above map is contained in $f + \mathfrak{m}^\perp$.*

Proof. For any $x \in \mathfrak{n}$ and $y \in \mathfrak{m}^\perp$, we have $\exp(\operatorname{ad} x)(f + y) \in f + \mathfrak{m}^\perp$, because $[\mathfrak{n}, \mathfrak{m}] \subset \mathfrak{m}$ and $\chi([\mathfrak{n}, \mathfrak{m}]) = 0$. Hence for any $m \in \mathfrak{m}$,

$$\begin{aligned}\big(\exp(\operatorname{ad} x)(f + y) - f \big| m\big) &= -(f \big| [f, m]) + \big(\exp(\operatorname{ad} x)(f + y) \big| m\big) \\ &= 0 + \big(\exp(\operatorname{ad} x)(f) \big| m\big) + \big(\exp(\operatorname{ad} y) \big| m\big),\end{aligned}$$

and since $\mathfrak{n}$ is ad-nilpotent,

$$\begin{aligned}&= \big(f \big| \exp(\operatorname{ad}(-x))(m)\big) + 0 \\ &= 0.\end{aligned}$$

This suffices to conclude, because $\mathbf{N}$ is generated by all $\exp(\operatorname{ad} x)$ with x running over $\mathfrak{n}$. Hence, by restriction, we obtain a map

$$\alpha \colon \mathbf{N} \times \mathcal{S}_f \longrightarrow f + \mathfrak{m}^\perp.$$

□

From [GG02, § 2.3] we have

Theorem 3.21. *The map α is an isomorphism of affine algebraic varieties.*

Consequently, the induced pullback

$$\alpha^* \colon \mathbb{C}[f + \mathfrak{m}^\perp] \longrightarrow \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]$$

is an isomorphism. Consider the α-equivariant action of $\mathbf{N}$ on $\mathbf{N} \times \mathcal{S}_f$ and on $f + \mathfrak{m}^\perp$, where the action on $\mathbf{N} \times \mathcal{S}_f$ is given by left translation on the first factor, i.e.,

$$n \cdot (g, s) := (ng, s), \qquad n, g \in \mathbf{N},\ s \in \mathcal{S}_f,$$

and the action on $f + \mathfrak{m}^\perp$ is the adjoint action, that is,

$$n \cdot (g \cdot s) = (ng) \cdot s \qquad \text{for all } n, g \in \mathbf{N},\ s \in \mathcal{S}_f.$$

Since α is $\mathbf{N}$-equivariant, we have

$$\mathbb{C}[f + \mathfrak{m}^\perp]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N} \times \mathcal{S}_f]^{\mathbf{N}} \cong \mathbb{C}[\mathbf{N}]^{\mathbf{N}} \otimes \mathbb{C}[\mathcal{S}_f],$$

and $\mathbb{C}[\mathbf{N}]^{\mathbf{N}} \cong \mathbb{C}$. This yields:

$$\mathbb{C}[\mathcal{S}_f] \cong \mathbb{C}[f + \mathfrak{m}^\perp]^{\mathbf{N}}.$$

Later, we will identify $\mathfrak{m}^\perp$ with $\mathfrak{m}^\circ := \{\xi \in \mathfrak{g}^* \mid \xi(m) = 0\}$ via $(\ |\ )$.

Now let us explicitly describe the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ (see [CG10, Proposition 1.3.18]). Let $(x_1, \ldots, x_n)$ be a basis of $\mathfrak{g}$ with $[x_i, x_j] = \sum_k c_{i,j}^k x_k$. Through the canonical isomorphism $(\mathfrak{g}^*)^* \cong \mathfrak{g}$, any element of $\mathfrak{g}$ can be regarded as a linear function on $\mathfrak{g}^*$, hence an element of $\mathbb{C}[\mathfrak{g}^*]$. For $f, g \in \mathbb{C}[\mathfrak{g}^*]$, we have

$$\{f, g\} = \sum_{i,j,k} c_{i,j}^k x_k \frac{\partial f}{\partial x_i^*} \frac{\partial g}{\partial x_j^*}.$$

More concisely, we have:

$$\{f, g\} \colon \mathfrak{g}^* \to \mathbb{C}, \quad \xi \mapsto \langle \xi, [d_\xi f, d_\xi g] \rangle, \tag{3.ii}$$

where $d_\xi f, d_\xi g \in (\mathfrak{g}^*)^* \cong \mathfrak{g}$ denote the differentials of f and g at $\xi \in \mathfrak{g}^*$. In particular, if $x, y \in \mathfrak{g} \cong (\mathfrak{g}^*)^* \subset \mathbb{C}[\mathfrak{g}^*]$, then

$$\{x, y\} = [x, y]. \tag{3.iii}$$

Moreover, for any regular functions $f, g \in \mathbb{C}[\mathfrak{g}^*]$ and any coadjoint orbit $\mathbb{O} \subset \mathfrak{g}^*$, we have

$$\{f, g\}|_{\mathbb{O}} = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}.$$

The bracket on the right comes from the symplectic structure on $\mathbb{O}$, while the bracket on the left is the restriction of the Poisson bracket on $\mathbb{C}[\mathfrak{g}^*]$ to $\mathbb{O}$ (see [CG10, Proposition 1.3.21]).

Definition 3.22. If X is smooth, one can view it as a complex-analytic manifold equipped with a holomorphic Poisson structure. For each point $x \in X$, the *symplectic leaf* $\mathcal{S}_x$ through x is defined as the set of points that can be reached from x by flowing along Hamiltonian vector fields.

Example 3.23. The space $\mathfrak{g}^*$ is a (smooth) Poisson variety. The Hamiltonian vector fields are given by all $\{f, \cdot\}$ with $f \in \mathbb{C}[\mathfrak{g}^*]$. Moreover, the symplectic leaves of $\mathfrak{g}^*$ are precisely the coadjoint orbits of $\mathfrak{g}^*$ (see [Vai94, Proposition 3.1]).

By a result of Weinstein (see [Wei83]), the natural Kirillov–Kostant–Souriau Poisson structure on $\mathfrak{g}^*$ can be induced on $\mathcal{S}_f$, i.e., the bracket $\{\ ,\ \}_{\mathcal{S}_f}$ on $\mathbb{C}[\mathcal{S}_f]$ is given by

$$\{f, g\}_{\mathcal{S}_f}(\xi) = \{f|_{\mathbb{O}}, g|_{\mathbb{O}}\}_{\text{symplectic}}(\xi), \tag{3.iv}$$

where for any $f, g \in \mathbb{C}[\mathcal{S}_f]$ and $\xi \in \mathcal{S}_f$, $\mathbb{O}$ denotes the coadjoint orbit in $\mathfrak{g}^*$ through ξ.

However, for the convenience of quantization, we introduce another method of obtaining the Poisson structure via Hamiltonian reduction.

Definition 3.24. An action of a Lie algebra $\mathfrak{a}$ on a manifold X is called *Hamiltonian* if there exists a Lie algebra homomorphism

$$\tilde{\mu}\colon \mathfrak{a} \longrightarrow \mathcal{O}_X(X), \qquad a \longmapsto \tilde{\mu}_a,$$

such that the following diagram commutes:

$$\begin{tikzcd} \mathfrak{a} \arrow[r] \arrow[dr, dashed, "\tilde{\mu}"'] & \mathfrak{X}(X) \\ & \mathcal{O}_X(X), \arrow[u] \end{tikzcd}$$

where $\mathfrak{X}(X)$ denotes the Lie algebra of symplectic vector fields on X, the vertical arrow is the natural embedding $\mathcal{O}_X(X) \hookrightarrow \mathfrak{X}(X)$ given by $f \mapsto \xi_f = \{f, \cdot\}$, which is a derivation on $\mathcal{O}_X(X)$, and the horizontal map $\mathfrak{a} \to \mathfrak{X}(X)$ is induced by the action of $\mathfrak{a}$ on X, specifically:

$$a \longmapsto (x \longmapsto \text{infinitesimal action in direction } a \curvearrowright x \in T_xX)\,.$$

The map $\tilde{\mu}$ is called the *co-moment map* for the action, or a *Hamiltonian* for the action. Its dual map

$$\mu\colon X \longrightarrow \mathfrak{a}^*, \qquad x \longmapsto \mu(x),$$

defined by $\langle \mu(x), a\rangle = \tilde{\mu}_a(x)$ for all $a \in \mathfrak{a}$, is called the *moment map* for the action.

Remark 3.25. If the group $\mathbf{A}$ is connected, then μ is $\mathbf{A}$-equivariant with respect to the coadjoint action on $\mathfrak{a}^*$.

Definition 3.26. Let X be a Poisson variety, $Y \subset X$ a subvariety, $\iota\colon Y \hookrightarrow X$ the inclusion, and $\pi\colon Y \twoheadrightarrow Z$ a surjective morphism. If there exists a Poisson structure $\{\,,\,\}_Z$ on Z such that for any open set $U \subset X$ and any

$$f, g \in \mathcal{O}_Z\big(\pi(U \cap Y)\big),$$

we have

$$\{f, g\}_Z \circ \pi(u) = \{\tilde{f}, \tilde{g}\} \circ \iota(u), \qquad \forall\, u \in U \cap Y,$$

where $\tilde{f}, \tilde{g} \in \mathcal{O}_X(U)$ are arbitrary extensions of

$$f \circ \pi|_{U \cap Y}, \qquad g \circ \pi|_{U \cap Y}$$

to U, then the triple (X, Y, Z) is called Poisson reducible.

The following theorem plays a crucial role in Hamiltonian reduction (see [LPV13, Proposition 5.39]):

Theorem 3.27 (Marsden–Weinstein)**.** *Let* $\mathbf{A}$ *be a connected algebraic group,* X *a Poisson variety, and suppose the action of* $\mathbf{A}$ *on* X *is Hamiltonian. Let*

$$\mu \colon X \to \mathfrak{a}^*$$

be the moment map, and let $\zeta \in \mathfrak{a}^*$ *be a regular value of* μ*. If the quotient*

$$\mu^{-1}(\zeta)/\mathbf{A}$$

exists as an algebraic variety, then $\mu^{-1}(\zeta)$ *is an* $\mathbf{A}$*-stable subvariety; denote by*

$$\iota \colon \mu^{-1}(\zeta) \hookrightarrow X, \qquad \pi \colon \mu^{-1}(\zeta) \twoheadrightarrow \mu^{-1}(\zeta)/\mathbf{A}$$

the inclusion and the quotient map. Then the triple

$$\big(X, \mu^{-1}(\zeta), \mu^{-1}(\zeta)/\mathbf{A}\big)$$

is Poisson reducible. In particular, $\mu^{-1}(\zeta)/\mathbf{A}$ *naturally inherits a Poisson structure.*

Remark 3.28. The Poisson structure obtained in this theorem is said to be obtained by Hamiltonian reduction.

Assume $\mathfrak{m} = \mathfrak{n}$, and in this case let $\mathbf{M}$ be the unipotent subgroup of $\mathbf{G}$ whose Lie algebra is $\mathfrak{m}$. The connected unipotent group $\mathbf{M}$ acts on the algebraic variety $\mathfrak{g}^*$ by the coadjoint action, and the algebra $\mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*)$ is equipped with the Lie–Poisson bracket. We intend to apply the above theorem to $\mathbf{M}$.

Proposition 3.29. *The above action of* $\mathbf{M}$ *on* $\mathfrak{g}^*$ *is Hamiltonian, and the moment map*

$$\mu\colon \mathfrak{g}^* \to \mathfrak{m}^*$$

is the restriction of linear functions on $\mathfrak{g}$ *to* $\mathfrak{m}$.

Proof. For a regular function $f : \mathfrak{g}^* \to \mathbb{C}$, its differential at $\xi \in \mathfrak{g}^*$ is the linear function

$$df_\xi : T_\xi(\mathfrak{g}^*) \to \mathbb{C}, \qquad df_\xi(v) = \left.\frac{d}{dt}\right|_{t=0} f(\xi + tv).$$

Since $\mathfrak{g}^*$ is a vector space, we have the isomorphism

$$T_\xi(\mathfrak{g}^*) \cong \mathfrak{g}^*,$$

and thus

$$df_\xi \in (\mathfrak{g}^*)^* \cong \mathfrak{g}.$$

In particular, for the linear function $f_x(\xi) := \xi(x)$ where $x \in \mathfrak{g}$, we have

$$df_{x,\xi} = x.$$

Consider the coadjoint action of M on $\mathfrak{g}^*$. Define

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad \tilde{\mu}_x(\xi) := \xi(x), \quad x \in \mathfrak{m},\ \xi \in \mathfrak{g}^*,$$

which means $d\tilde{\mu}_{x,\xi} = x$. Then

$$\tilde{\mu} : \mathfrak{m} \to \mathcal{O}_{\mathfrak{g}^*}(\mathfrak{g}^*), \qquad x \mapsto \tilde{\mu}_x$$

is a co-moment map. Indeed, by Equation 3.ii, for $x, y \in \mathfrak{m}$,

$$\{\tilde{\mu}_x, \tilde{\mu}_y\}(\xi) = \xi([x, y]) = \tilde{\mu}_{[x,y]}(\xi),$$

so $\tilde{\mu}$ is a Lie algebra homomorphism. By definition, the corresponding moment map

$$\mu : \mathfrak{g}^* \to \mathfrak{m}^*$$

is the dual map characterized by

$$\mu(\xi)(x) = \tilde{\mu}_x(\xi) = \xi(x), \qquad x \in \mathfrak{m}.$$

Therefore

$$\mu(\xi) = \xi|_{\mathfrak{m}}.$$

We now show that the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian. It remains to verify that $\tilde{\mu}_x$ generates the infinitesimal coadjoint action induced by x. For $y \in \mathfrak{g} \cong (\mathfrak{g}^*)^*$, making no distinction between y and f_y, we have by Equation 3.iii:

$$\begin{aligned}\left(\frac{d}{dt}\bigg|_{t=0} \exp(t\,\mathrm{ad}_x^*)\xi\right)(y) &= \frac{d}{dt}\bigg|_{t=0} \big((\exp(t\,\mathrm{ad}_x^*)\xi)(y)\big) \\ &= (\mathrm{ad}_x^*\,\xi)(y) \\ &= -\xi([x, y]).\end{aligned}$$

On the other hand,

$$\{\tilde{\mu}_x, f_y\}(\xi) = \xi([x, y]).$$

Therefore the Hamiltonian vector field of $\tilde{\mu}_x$ coincides with the infinitesimal vector field induced by x. Hence the coadjoint action of M on $\mathfrak{g}^*$ is Hamiltonian. $\square$

Since $\chi|_{\mathfrak{m}}$ is a character on $\mathfrak{x}$, we have

$$\big(\mathrm{ad}^*\, m(\chi)\big)y = -\chi([x, y]) = 0 \quad x, y \in \mathfrak{m}.$$

Hence

$$\mathrm{ad}^* x(\chi|_{\mathfrak{m}}) = 0.$$

That is,

$$\mathrm{Ad}^* m(\chi|_{\mathfrak{m}}) = \chi|_{\mathfrak{m}} \quad m \in M.$$

Thus the set

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \{\xi \in \mathfrak{g}^* \mid \mu(\xi) = \chi|_{\mathfrak{m}}\}$$

is **M**-stable. Moreover,

$$\mu(\xi) = \chi|_{\mathfrak{m}} \iff (\xi - \chi)|_{\mathfrak{m}} = 0 \iff \xi - \chi \in \mathfrak{m}^\circ \cong \mathfrak{m}^\perp.$$

Hence,

$$\mu^{-1}(\chi|_{\mathfrak{m}}) = \chi + \mathfrak{m}^\perp.$$

Furthermore, we have the following lemma:

Lemma 3.30 ([AM16, Lemma 3.23])**.** *For every symplectic leaf of $\mathfrak{g}^*$, $\chi|_{\mathfrak{m}}$ is a regular value of the restriction of μ.*

Since the map

$$\mathbf{M} \times \mathcal{S}_f \longrightarrow \chi + \mathfrak{m}^\perp$$

is an isomorphism of affine algebraic varieties,

$$\mathcal{S}_f \cong (\chi + \mathfrak{m}^\perp)/\mathbf{M} = \mu^{-1}(\chi|_{\mathfrak{m}})/\mathbf{M}.$$

Thus the conditions of Theorem 3.27 are satisfied. We obtain a Poisson structure on $\mathcal{S}_f$. Its description is as follows:

Let $\pi\colon \chi + \mathfrak{m}^\perp \twoheadrightarrow (\chi + \mathfrak{m}^\perp)/\mathbf{M} \simeq \mathcal{S}_f$ be the natural projection, and $\iota\colon \chi + \mathfrak{m}^\perp \hookrightarrow \mathfrak{g}^*$ the natural inclusion. Then for any $f, g \in \mathbb{C}[\mathcal{S}_f]$,

$$\{f, g\}_{\mathcal{S}_f} \circ \pi = \{\tilde{f}, \tilde{g}\} \circ \iota$$

where $\tilde{f}, \tilde{g}$ are arbitrary extensions of $f \circ \pi, g \circ \pi$ to $\mathfrak{g}^*$. One can see that the Poisson structure defined on $\mathcal{S}_f$ by the latter coincides with the one defined in Equation 3.iv. Set

$$I_\chi = \mathbb{C}[\mathfrak{g}^*] \sum_{x \in \mathfrak{m}} (x - \chi(x))$$

where $\mathbb{C}[\mathfrak{g}^*]$ is identified with $\mathbb{C}[\mathfrak{g}]$, so

$$\mathbb{C}[\mu^{-1}(\chi)] = \mathbb{C}[\mathfrak{g}^*]/I_\chi.$$

Then $\mathbb{C}[\mathcal{S}_f] = \mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ can be identified with the subspace of $\mathbb{C}[\mathfrak{g}^*]/I_\chi$ consisting of all cosets $\phi + I_\chi$ such that for every $x \in \mathfrak{m}$, $\{x, \phi\} \in I_\chi$. Under this realization, the Poisson structure on $\mathbb{C}[\mathcal{S}_f]$ is defined by

$$\{\phi + I_\chi, \phi' + I_\chi\} = \{\phi, \phi'\} + I_\chi$$

for ϕ, ϕ' satisfying $\{x, \phi\}, \{x, \phi'\} \in I_\chi$ for all $x \in \mathfrak{m}$.

Example 3.31. Write a general element of $\mathfrak{sl}_4$ as

$$X = (x_{ij})_{1 \le i,j \le 4}, \qquad \sum_{i=1}^{4} x_{ii} = 0.$$

Because

$$\mathfrak{m} = \text{span}\{E_{12}, E_{14}, E_{23}, E_{43}, E_{13}\},$$

and $\chi + \mathfrak{m}^\perp \in \mathfrak{g}^*$ is isomorphic to $f + \mathfrak{m}^\perp$, a general point in $\chi + m^\perp$ has the form

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ 2 & x_{22} & x_{23} & x_{24} \\ 0 & 2 & x_{33} & 0 \\ 0 & x_{42} & x_{43} & x_{44} \end{pmatrix}, \qquad x_{44} = -(x_{11} + x_{22} + x_{33}).$$

Unfortunately, I am unable to compute an explicit expression for $\mathbb{C}[\mu^{-1}(\chi)]^{\mathbf{M}}$ in the $\mathfrak{sl}_4$ case or any nontrivial case. Let us briefly explain why the computation

is so difficult. Recall that $\alpha : M \times S_f \longrightarrow f + m^{\perp}$ is an isomorphism of affine algebraic varieties. Recalling the whole process, we have

$$(\mathbb{C}[\mathfrak{g}^*]/I_\chi)^M \longrightarrow \mathbb{C}^M[\mu^{-1}(\chi|_m)] \longrightarrow \mathbb{C}^M[f+m^{\perp}] \longrightarrow \mathbb{C}^M[M\times S_f] \longrightarrow \mathbb{C}[S_f].$$

We can choose suitable representatives G so that, at the level of elements, the above map can be written as

$$\overline{G} \longmapsto G \longmapsto G\circ k \longmapsto G\circ k\circ\alpha \longmapsto \sum (G\circ k\circ\alpha)_1 \otimes (G\circ k\circ\alpha)_2,$$

where

$$\begin{aligned} \kappa :\ f + m^{\perp} &\longrightarrow \chi + m^{\circ} \\ F &\longmapsto (F|\ ), \end{aligned}$$

and the map between $\mathbb{C}^M[M \times S_f]$ and $\mathbb{C}[S_f]$ at the element level is given by

$$\begin{aligned} \mathbb{C}^M[M \times S_f] &\xrightarrow{\sim} \mathbb{C}^M[M] \otimes \mathbb{C}[S_f] \\ F &\longmapsto \sum F_1 \otimes F_2 \quad \text{s.t. } F(m,s) = \sum F_1(m)F_2(s). \end{aligned}$$

It is easy to see that characterizing the elements in $\mathbb{C}^M[M \times S_f]$ is difficult, and the map between it and $\mathbb{C}[S_f]$ is exceedingly hard to describe explicitly in general. Since all the above arrows are isomorphisms, it is natural to consider the reverse arrows, and we will now explain that the obstacles cannot be avoided either.

$$\mathbb{C}[S_f] \longrightarrow \mathbb{C}^M[M\times S_f] \longrightarrow \mathbb{C}^M[f+m^{\perp}] \longrightarrow \mathbb{C}^M[\mu^{-1}(\chi|_{\mathfrak{m}})] \longrightarrow (\mathbb{C}[\mathfrak{g}^*]/I_\chi)^M.$$

At the element level we have

$$G \longmapsto (1,G) \longmapsto (1,G)\cdot\alpha^{-1} \longmapsto (1,G)\cdot\alpha^{-1}\cdot k^{-1} \longmapsto \overline{(1,G)\cdot\alpha^{-1}\cdot k^{-1}},$$

where

$$\kappa^{-1}:\ \chi + m^{\circ} \longrightarrow f + m^{\perp}$$

$$H(\cdot) = (f_H|\cdot) \longmapsto f_H.$$

It is easy to see that for any given element in $\mu^{-1}(\chi|_{\mathfrak{m}})$, its image under α^{-1} can hardly be determined; moreover, the image under κ^{-1} is equally hard to describe. Exactly for this reason, we cannot give generators and relations for this Poisson structure in concrete examples. Nevertheless, we can still proceed to discuss the quantization of the Slodowy slice.

Chapter 4

Quantization of the Slodowy Slice and Finite W-Algebras

In this chapter we will discuss BRST cohomology and the quantization of the Slodowy slice. In Section 4.1, we define BRST cohomology by means of a specific graded Clifford algebra, and finally show that this cohomology realizes the Poisson structure on the Slodowy slice. In Section 4.2, we present the quantized version of BRST cohomology and show that it provides a quantization of the Slodowy slice. In Section 4.3, we give the classical definition of finite W-algebras and show that it coincides with the quantization of the Slodowy slice. The content of this chapter is expounded in detail in [Ara17], and we follow its notations.

4.1 BRST Realization of the Slodowy Slice

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra, G its connected affine algebraic group. Let $\{x_i\}_{1\leq i\leq d}$ be a basis of $\mathfrak{g}$, and $\{x_i^*\}_{1\leq i\leq d}$ the corresponding dual basis of $\mathfrak{g}^*$. For $i, j = 1, \ldots, d$, we have $[x_i, x_j] = \sum_{k=1}^{d} c_{i,j}^k x_k$. Let $Cl(\mathfrak{g})$ be the *Clifford algebra* associated with the finite-dimensional vector space $\mathfrak{g} \oplus \mathfrak{g}^*$ and the nondegenerate

bilinear form $B(\cdot,\cdot) := \langle\cdot|\cdot\rangle$, where

$$\langle f + x \mid g + y\rangle = f(y) + g(x)$$

for $f, g \in \mathfrak{g}^*$, $x, y \in \mathfrak{g}$. Thus the super Lie bracket (superbracket) is given as follows:

For homogeneous elements $a, b \in Cl(\mathfrak{g})$,

$$[a, b] = ab - (-1)^{|a||b|}ba.$$

A direct computation yields the Leibniz rule for homogeneous $a, b, c \in Cl(\mathfrak{g})$:

$$[a, bc] = [a, b]c + (-1)^{|a||b|}b[a, c].$$

In particular,

$$[x, f] = f(x) \quad x \in \mathfrak{g} \subset \wedge\mathfrak{g},\ f \in \mathfrak{g}^* \subset \wedge\mathfrak{g}^*.$$

$Cl(\mathfrak{g})$ is a unital superalgebra. From Equation 2.i and [Mei13, § 2.1.2] we know that, as a $\mathbb{C}$-linear space, it is isomorphic to

$$\wedge(\mathfrak{g} \oplus \mathfrak{g}^*) \cong \wedge(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*),$$

and the natural embeddings

$$\wedge(\mathfrak{g}) \hookrightarrow Cl(\mathfrak{g}), \qquad \wedge(\mathfrak{g}^*) \hookrightarrow Cl(\mathfrak{g})$$

are homogeneous homomorphisms of superalgebras.

Define an increasing filtration on $Cl(\mathfrak{g})$ by

$$Cl_p(\mathfrak{g}) := \wedge^{\leqslant p}(\mathfrak{g}) \otimes \wedge(\mathfrak{g}^*).$$

We have

$$0 = Cl_{-1}(\mathfrak{g}) \subset Cl_0(\mathfrak{g}) \subset Cl_1(\mathfrak{g}) \subset \cdots \subset Cl_d(\mathfrak{g}) = Cl(\mathfrak{g}),$$

and

$$Cl_p(\mathfrak{g}) \cdot Cl_q(\mathfrak{g}) \subset Cl_{p+q}(\mathfrak{g}), \qquad [Cl_p(\mathfrak{g}), Cl_q(\mathfrak{g})] \subset Cl_{p+q-1}(\mathfrak{g}),$$

where the first inclusion is easy to prove. To verify the second, take $x \in Cl_p(\mathfrak{g}) \setminus Cl_{p-1}(\mathfrak{g})$ and $y \in Cl_q(\mathfrak{g}) \setminus Cl_{q-1}(\mathfrak{g})$; we only need to verify $[x, y] \in Cl_{q+p-1}(\mathfrak{g})$, which follows by a direct computation from the definitions.

Let $\overline{Cl}(\mathfrak{g})$ be the associated graded algebra:

$$\overline{Cl}(\mathfrak{g}) := \operatorname{gr} Cl(\mathfrak{g}) = \bigoplus_{p \geqslant 0} \frac{Cl_p(\mathfrak{g})}{Cl_{p-1}(\mathfrak{g})}.$$

Then $\overline{Cl}(\mathfrak{g})$ has the structure of a graded Poisson superalgebra with respect to the super Lie bracket, called the *classical Clifford algebra* associated with $\mathfrak{g} \oplus \mathfrak{g}^*$. Its Poisson superbracket is given by

$$\{x, f\} = f(x), \qquad x \in \mathfrak{g} \subset \wedge(\mathfrak{g}), \quad f \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*),$$

$$\{x, y\} = 0, \qquad x, y \in \mathfrak{g} \subset \wedge(\mathfrak{g}),$$

$$\{f, g\} = 0, \qquad f, g \in \mathfrak{g}^* \subset \wedge(\mathfrak{g}^*).$$

Remark 4.1. This suffices to define the whole super Poisson structure. We illustrate the calculation with an example.

Example 4.2. Take

$$A = x_a x_b x_c^* \in Cl_2(\mathfrak{g}), \qquad B = x_d x_e^* \in Cl_1(\mathfrak{g}).$$

Let us compute $\{A, B\}$ using the above relations and Definition 2.26. Since A is an odd element, by the super Leibniz rule we obtain

$$\{x_a x_b x_c^*, x_d x_e^*\} = \{x_a x_b x_c^*, x_d\} x_e^* - x_d \{x_a x_b x_c^*, x_e^*\}.$$

where

$$\begin{aligned}\{x_a x_b x_c^*, x_d\} &= x_a x_b \{x_c^*, x_d\} - \{x_a x_b, x_d\} x_c^* \\ &= \delta_{cd} x_a x_b.\end{aligned}$$

and

$$\begin{aligned}\{x_a x_b x_c^*, x_e^*\} &= x_a x_b \{x_c^*, x_e^*\} - \{x_a x_b, x_e^*\} x_c^*. \\ &= -\delta_{be} x_a x_c^* + \delta_{ae} x_b x_c^*.\end{aligned}$$

Substituting back,

$$\begin{aligned}\{x_a x_b x_c^*, x_d x_e^*\} &= \delta_{cd} x_a x_b x_e^* - x_d \left(-\delta_{be} x_a x_c^* + \delta_{ae} x_b x_c^*\right) \\ &= \delta_{cd} x_a x_b x_e^* + \delta_{be} x_d x_a x_c^* - \delta_{ae} x_d x_b x_c^*. \\ &= \delta_{cd} x_a x_b x_e^* - \delta_{be} x_a x_d x_c^* + \delta_{ae} x_b x_d x_c^*.\end{aligned}$$

In particular, one easily finds

$$\{x_a x_b x_c^*, x_d x_e^*\} \in Cl_2(\mathfrak{g}).$$

This is consistent with

$$\{Cl_p(\mathfrak{g}), Cl_q(\mathfrak{g})\} \subset Cl_{p+q-1}(\mathfrak{g})$$

for $p = 2, q = 1$.

Let $\bar{C}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g})$. Then $\bar{C}(\mathfrak{g})$ is naturally a Poisson superalgebra, as it is the tensor product of Poisson superalgebras, where the superalgebra structure on $\mathbb{C}[\mathfrak{g}^*]$ is given by the polynomial grading modulo 2.

Direct verification yields the following (see [AM24, Lemma 7.1]):

Lemma 4.3. *The following map gives a Lie algebra homomorphism:*

$$\rho : \mathfrak{g} \to Cl(\mathfrak{g}), \quad x_i \mapsto \sum_{1 \le j,k \le d} c_{i,j}^k x_k x_j^*,$$

and for $x, y \in \mathfrak{g}$*,*

$$[\rho(x), y] = [x, y] \in \mathfrak{g} \subset Cl(\mathfrak{g}),$$

where the first bracket is the Lie bracket in $Cl(\mathfrak{g})$*, and the second is the Lie bracket in* $\mathfrak{g}$*.*

Remark 4.4. $\rho : \mathfrak{g} \to Cl_1(\mathfrak{g}) \subset Cl(\mathfrak{g})$ naturally induces a Lie algebra homomorphism

$$\bar{\rho} := \sigma_1 \circ \rho : \mathfrak{g} \to \bar{C}l(\mathfrak{g}),$$

where σ_1 is the projection $Cl_1(\mathfrak{g}) \to Cl_1(\mathfrak{g})/Cl_0(\mathfrak{g}) \subset \operatorname{gr} Cl(\mathfrak{g})$.

Lemma 4.5. *The following map gives a Lie algebra homomorphism:*

$$\begin{aligned} \bar{\theta}_\chi : \mathfrak{g} &\longrightarrow \bar{C}(\mathfrak{g}) \\ x &\longmapsto (x - \chi(x)) \otimes 1 + 1 \otimes \bar{\rho}(x). \end{aligned}$$

Proof. A direct computation using the fact that χ is a character on $\mathfrak{g}$ yields the result. □

Let

$$\bar{C}(\mathfrak{g}) = \bigoplus_{n \in \mathbb{Z}} \bar{C}^n(\mathfrak{g})$$

be the $\mathbb{Z}$-grading defined by:

$$\bar{C}^n(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \left(\bigoplus_{j-i=n} \wedge^i(\mathfrak{g}) \otimes \wedge^j(\mathfrak{g}^*) \right).$$

with

$$\deg(\phi \otimes 1) = 0 \quad \phi \in \mathbb{C}[\mathfrak{g}^*],$$

$$\deg(1 \otimes f) = 1 \quad f \in \mathfrak{g}^*,$$

$$\deg(1 \otimes x) = -1 \quad x \in \mathfrak{g}.$$

Lemma 4.6. *There exists a unique element $\bar{D} \in \bar{C}^1(\mathfrak{g})$ such that*

$$\{\bar{D}, 1 \otimes x\} = \bar{\theta}_\chi(x) \qquad \textit{for } x \in \mathfrak{g}.$$

Moreover,

$$\{\bar{D}, \bar{D}\} = 0.$$

Proof. First note that

$$\overline{Cl}(\mathfrak{g})^{\mathfrak{g}} := \{w \in \overline{Cl}(\mathfrak{g}) \mid \{x, w\} = 0,\ \forall x \in \mathfrak{g}\} = \wedge(\mathfrak{g}).$$

Set

$$\bar{D} = \sum_i \big(x_i - \chi(x_i)\big) \otimes x_i^* - 1 \otimes \frac{1}{2} \sum_{i,j,k} c_{i,j}^k x_i^* x_j^* x_k.$$

Using the super Leibniz rule and

$$\{x_i^*, x_j\} = \delta_{ij}, \qquad \{x_i, x_j\} = 0,$$

a direct computation gives

$$\{\bar{D}, 1 \otimes x\} = (x - \chi(x)) \otimes 1 + 1 \otimes \bar{\rho}(x) = \bar{\theta}_\chi(x),$$

which proves existence.

Now prove uniqueness. If $\bar{D}_1, \bar{D}_2 \in \overline{C}^1(\mathfrak{g})$ both satisfy the property, set

$$R = \bar{D}_1 - \bar{D}_2.$$

Then

$$\{R, 1 \otimes x\} = 0, \qquad \forall x \in \mathfrak{g}.$$

Hence

$$R \in \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g})^{\mathfrak{g}} = \mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g}).$$

By the definition of degree, elements in $\big(\mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g})\big)$ have degree at most 0, so

$$\big(\mathbb{C}[\mathfrak{g}^*] \otimes \wedge(\mathfrak{g})\big) \cap \overline{C}^1(\mathfrak{g}) = 0.$$

Thus $R = 0$, which proves uniqueness.

Finally, from the Jacobi superidentity together with

$$\{\bar{D}, 1 \otimes x\} = \bar{\theta}_\chi(x)$$
$$[\overline{\rho}(x), y] = [x, y],$$

one can directly compute that for any $x, y \in \mathfrak{g}$,

$$\{1 \otimes x, \{1 \otimes y, \{\bar{D}, \bar{D}\}\}\} = 0.$$

Combined with the conclusion $\overline{Cl}(\mathfrak{g})^{\mathfrak{g}} = \wedge(\mathfrak{g})$ and degree considerations, we get

$$\{\bar{D}, \bar{D}\} = 0.$$

□

Consider

$$\operatorname{ad} \bar{D} := \{\bar{D}, \cdot\}.$$

By Lemma 4.6 and the Jacobi superidentity, this satisfies

$$(\operatorname{ad} \bar{D})^2 = 0.$$

Therefore,

$$(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

is a *differential graded Poisson superalgebra*. Its cohomology

$$H^\bullet_{BRST,\chi}(\mathfrak{g}, \mathbb{C}[\mathfrak{g}^*]) := H^\bullet(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D}) = \bigoplus_{i \in \mathbb{Z}} H^i(\overline{C}(\mathfrak{g}), \operatorname{ad} \bar{D})$$

inherits from $\overline{C}(\mathfrak{g})$ a graded Poisson superalgebra structure.

More generally, let P be a Poisson algebra equipped with a Poisson algebra homomorphism

$$\mu^* : \mathbb{C}[\mathfrak{g}^*] \to P.$$

Since the map

$$\mu^* \otimes \mathrm{id} : \overline{C}(\mathfrak{g}) \to P \otimes \overline{Cl}(\mathfrak{g})$$

is a Poisson superalgebra homomorphism, we may regard $\bar{D}$ as an element of

$$P \otimes \overline{Cl}(\mathfrak{g}).$$

Now

$$(P \otimes \overline{Cl}(\mathfrak{g}), \mathrm{ad}\, \bar{D})$$

is a differential graded Poisson superalgebra. Hence its cohomology

$$H^{\bullet}_{BRST,\chi}(\mathfrak{g}, P) := H^{\bullet}(P \otimes \overline{Cl}(\mathfrak{g}), \mathrm{ad}\, \bar{D}) \tag{4.i}$$

inherits from

$$P \otimes \overline{Cl}(\mathfrak{g})$$

a graded Poisson superalgebra structure.

Let $\mathfrak{m}$ be as defined in Equation 3.i. Then we obtain the BRST realization of the Slodowy slice (see [Ara17, § 2.7]):

Theorem 4.7. *We have*

$$H^{i}_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) = 0 \qquad \textit{for } i \neq 0,$$

and

$$H^{0}_{BRST,\chi}(\mathfrak{m}, \mathbb{C}[\mathfrak{g}^*]) \cong \mathbb{C}[\mathcal{S}_f]$$

as Poisson algebras.

Next, we quantize the Slodowy slice by means of quantized BRST cohomology.

4.2 Quantized BRST Cohomology and Quantization of the Slodowy Slice

The ideas and conclusions of this section run completely parallel to those of the previous section. Let $\{U_i(\mathfrak{g})\}$ be the PBW filtration on the universal enveloping algebra $U(\mathfrak{g})$ of $\mathfrak{g}$. By the PBW theorem, there is a Poisson algebra isomorphism

$$\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{i\geq 0} U_i(\mathfrak{g})/U_{i-1}(\mathfrak{g}) \cong S(\mathfrak{g}) \cong \mathbb{C}[\mathfrak{g}^*].$$

Here the Poisson structure on $\mathbb{C}[\mathfrak{g}^*] \cong S(\mathfrak{g})$ is determined by

$$\{x, y\} = [x, y], \qquad x, y \in \mathfrak{g} \subset S(\mathfrak{g}),$$

and extended uniquely by the Leibniz rule.

Set

$$C(\mathfrak{g}) = U(\mathfrak{g}) \otimes Cl(\mathfrak{g}).$$

Here $U(\mathfrak{g})$ is regarded as a purely even superalgebra, so the $\mathbb{Z}_2$-grading on $C(\mathfrak{g})$ is entirely determined by that of $Cl(\mathfrak{g})$. Using the filtrations on $U(\mathfrak{g})$ and $Cl(\mathfrak{g})$, define a filtration on $C(\mathfrak{g})$ by

$$C_p(\mathfrak{g}) = \sum_{i+j\leq p} U_i(\mathfrak{g}) \otimes Cl_j(\mathfrak{g}).$$

Recall

$$\overline{C}(\mathfrak{g}) = \mathbb{C}[\mathfrak{g}^*] \otimes \overline{Cl}(\mathfrak{g}),$$

and we have a graded Poisson superalgebra isomorphism

$$\operatorname{gr} C(\mathfrak{g}) \cong \overline{C}(\mathfrak{g}).$$

Indeed, the map

$$\begin{aligned} \Phi :\ \operatorname{gr} U(\mathfrak{g}) \otimes \operatorname{gr} Cl(\mathfrak{g}) &\longrightarrow \operatorname{gr} C(\mathfrak{g}) \\ (u + U_{i-1}) \otimes (a + Cl_{j-1}) &\longmapsto u \otimes a + C_{i+j-1}. \end{aligned}$$

is an isomorphism of Poisson superalgebras. Hence $C(\mathfrak{g})$ can be viewed as a quantization of $\overline{C}(\mathfrak{g})$.

Define a $\mathbb{Z}$-grading on $C(\mathfrak{g})$ as follows:

$$C^n(\mathfrak{g}) = U(\mathfrak{g}) \otimes \left(\bigoplus_{j-i=n} \wedge^i(\mathfrak{g}) \otimes \wedge^j(\mathfrak{g}^*) \right).$$

where

$$\deg(u \otimes 1) = 0, \qquad u \in U(\mathfrak{g}),$$

$$\deg(1 \otimes f) = 1, \qquad f \in \mathfrak{g}^*,$$

$$\deg(1 \otimes x) = -1, \qquad x \in \mathfrak{g}.$$

Thus

$$C(\mathfrak{g}) = \bigoplus_{n \in \mathbb{Z}} C^n(\mathfrak{g}).$$

Recall the Lie algebra homomorphism defined earlier:

$$\rho : \mathfrak{g} \longrightarrow Cl(\mathfrak{g}), \qquad x_i \longmapsto \sum_{1 \leq j,k \leq d} c_{i,j}^k x_k x_j^*.$$

Likewise, we have:

Lemma 4.8. *The map*

$$\theta_\chi : \mathfrak{g} \longrightarrow C(\mathfrak{g}), \qquad x \longmapsto (x - \chi(x)) \otimes 1 + 1 \otimes \rho(x)$$

is a Lie algebra homomorphism.

Proof. Since $\rho : \mathfrak{g} \to Cl(\mathfrak{g})$ is a Lie algebra homomorphism and $\chi([\mathfrak{g}, \mathfrak{g}]) = 0$, a direct computation yields

$$[\theta_\chi(x), \theta_\chi(y)] = \theta_\chi([x, y]), \qquad x, y \in \mathfrak{g}.$$

□

Lemma 4.9. *There exists a unique element $D \in C^1(\mathfrak{g})$ such that for every $x \in \mathfrak{g}$,*

$$[D, 1 \otimes x] = \theta_\chi(x).$$

Moreover,

$$D^2 = 0.$$

Proof. First note that

$$Cl(\mathfrak{g})^{\mathfrak{g}} := \{w \in Cl(\mathfrak{g}) \mid [x, w] = 0,\ \forall x \in \mathfrak{g}\} = \wedge(\mathfrak{g}).$$

Set

$$D = \sum_i (x_i - \chi(x_i)) \otimes x_i^* - 1 \otimes \frac{1}{2} \sum_{i,j,k} c_{i,j}^k x_i^* x_j^* x_k.$$

Using the super Leibniz rule and

$$[x_i^*, x_j] = \delta_{ij}, \qquad [x_i, x_j] = 0,$$

a direct computation gives

$$[D, 1 \otimes x] = (x - \chi(x)) \otimes 1 + 1 \otimes \rho(x) = \theta_\chi(x).$$

Existence is proved.

The proof of uniqueness and $D^2 = 0$ follows the same method as in the previous section. □

Consider

$$\operatorname{ad} D = [D, -].$$

By the above lemma and the Jacobi superidentity, we have

$$(\operatorname{ad} D)^2 = 0.$$

Thus

$$(C(\mathfrak{g}), \operatorname{ad} D)$$

is a differential graded superalgebra. Its cohomology is denoted by

$$H^{\bullet}_{\mathrm{BRST},\chi}(\mathfrak{g}, U(\mathfrak{g})) := H^{\bullet}(C(\mathfrak{g}), \operatorname{ad} D) = \bigoplus_{i\in\mathbb{Z}} H^{i}(C(\mathfrak{g}), \operatorname{ad} D).$$

It inherits a graded superalgebra structure from $C(\mathfrak{g})$.

More generally, let A be a $U(\mathfrak{g})$-algebra. By the natural map

$$C(\mathfrak{g}) = U(\mathfrak{g}) \otimes Cl(\mathfrak{g}) \longrightarrow A \otimes Cl(\mathfrak{g}),$$

we still denote the image of D by D. Then

$$(A \otimes Cl(\mathfrak{g}), \operatorname{ad} D)$$

is a differential graded superalgebra. Hence we define

$$H^{\bullet}_{\mathrm{BRST},\chi}(\mathfrak{g}, A) := H^{\bullet}(A \otimes Cl(\mathfrak{g}), \operatorname{ad} D).$$

When $\chi = 0$, we simply write

$$H^{\bullet}_{\mathrm{BRST}}(\mathfrak{g}, A).$$

Now take $\mathfrak{m}$ as defined earlier. Define

$$U(\mathfrak{g}, f) := H^{0}_{\mathrm{BRST},\chi}(\mathfrak{m}, U(\mathfrak{g})).$$

This algebra is called the finite W-algebra associated with f. From the computation of classical BRST cohomology and the filtration isomorphism above, we have

$$\operatorname{gr} U(\mathfrak{g}, f) \cong \mathbb{C}[\mathcal{S}_f].$$

Therefore $U(\mathfrak{g}, f)$ is a quantization of the coordinate ring $\mathbb{C}[\mathcal{S}_f]$ of the Slodowy slice $\mathcal{S}_f$.

Finite W-algebras themselves are algebraic structures worthy of study, see for instance [BK06; Lu+25; AE22]. On the other hand, finite W-algebras also admit a more algebraic and more commonly used definition, namely through the Whittaker model.

4.3 Whittaker Model Realization of Finite W-Algebras

Let

$$\chi = (f \mid \cdot) \in \mathfrak{g}^*.$$

By restriction, χ defines a one-dimensional representation

$$\chi : U(\mathfrak{m}) \to \mathbb{C}.$$

We denote by $\mathbb{C}_\chi$ the corresponding left $U(\mathfrak{m})$-module.

Let I_χ be the left ideal of $U(\mathfrak{g})$ generated by the elements

$$x - \chi(x), \qquad x \in \mathfrak{m},$$

i.e.,

$$I_\chi := \sum_{x \in \mathfrak{m}} U(\mathfrak{g})(x - \chi(x)).$$

There is a natural isomorphism of left $U(\mathfrak{g})$-modules

$$Q_\chi := U(\mathfrak{g}) \otimes_{U(\mathfrak{m})} \mathbb{C}_\chi \cong U(\mathfrak{g})/I_\chi.$$

The $U(\mathfrak{g})$-module Q_χ is called the *generalized Gelfand–Graev module* associated with f.

The adjoint action of $\mathfrak{m}$ on $\mathfrak{g}$ extends to an action on $U(\mathfrak{g})$, and the ideal I_χ is $\mathfrak{m}$-stable, so the quotient Q_χ carries a natural $\mathfrak{m}$-module structure. We define

$$Q_\chi^{\mathfrak{m}} := \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi,\ \forall x \in \mathfrak{m}\},$$

where $\bar{u}$ denotes the image of $u \in U(\mathfrak{g})$ in Q_χ.

Definition 4.10. The finite W-algebra associated with f is defined as

$$U(\mathfrak{g}, f) := Q_\chi^{\mathfrak{m}}.$$

Equivalently,

$$U(\mathfrak{g}, f) = \{\bar{u} \in Q_\chi \mid [x, u] \in I_\chi,\ \forall x \in \mathfrak{m}\}.$$

This realization of $U(\mathfrak{g}, f)$ is called the *Whittaker model realization*.

Arakawa proved the following theorem in [Ara07a].

Theorem 4.11. *There is a natural algebra isomorphism*

$$U(\mathfrak{g}, f) \cong \mathrm{End}_{U(\mathfrak{g})}(Q_\chi)^{\mathrm{op}} \cong Q_\chi^{\mathfrak{m}}.$$

Here op *denotes the opposite algebra structure on* $\mathrm{End}_{U(\mathfrak{g})}(Q_\chi)$*, i.e., multiplication is given by reverse composition:*

$$\varphi \cdot \psi := \psi \circ \varphi.$$

Thus we have presented two very different realizations of W-algebras, one geometric and one algebraic. The above theorem shows that they are identical.

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